

CPU Test Platform Manual

Goldenbell1B Board

CRZ Technology

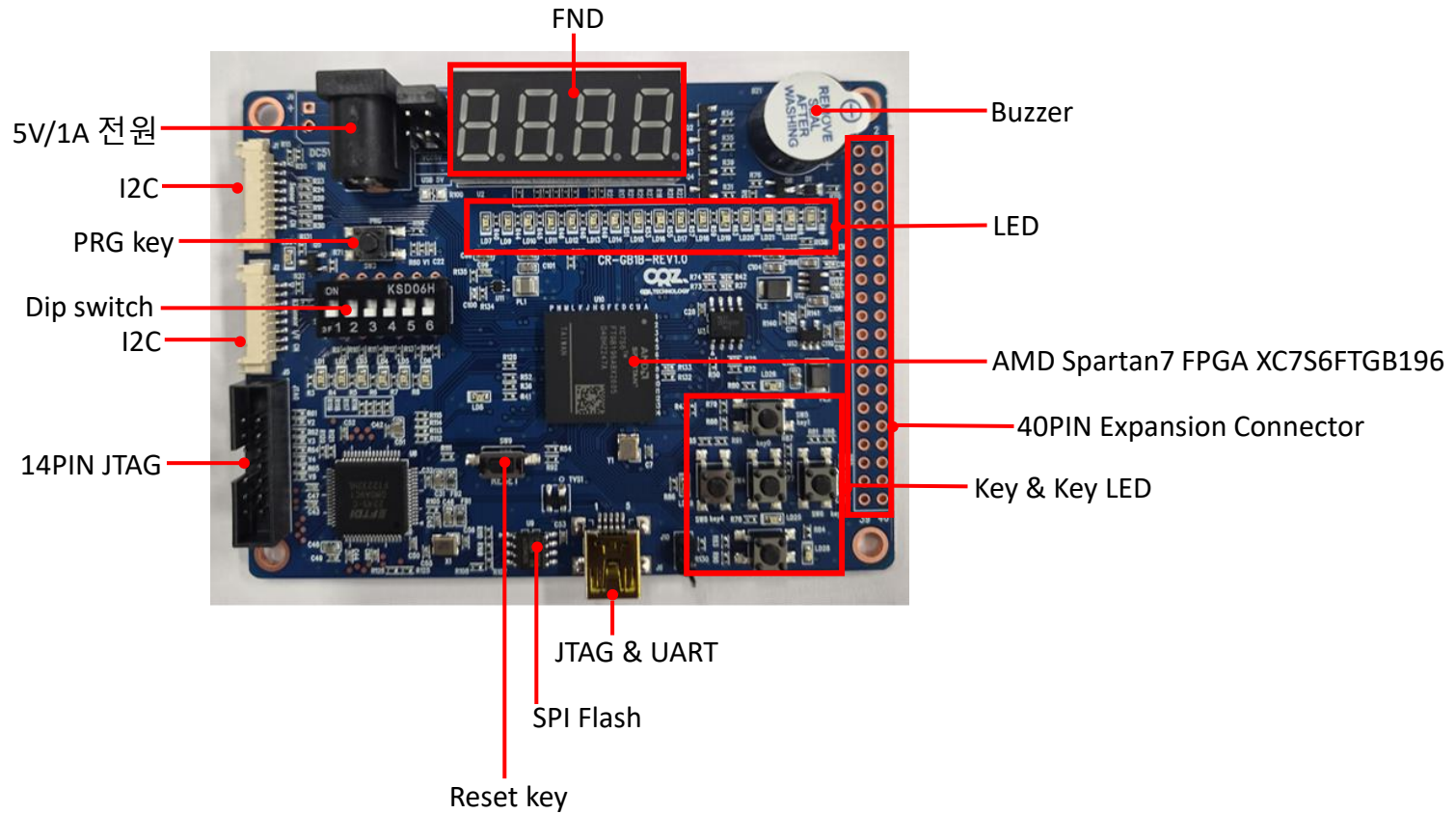
Goldenbell1B Board Features

- AMD Spartan7 FPGA (XC7S6FTGB196)
 - Vivado 2025.1
 - 아래 링크에서 Vivado 2025.1 download/install

https://www.xilinx.com/member/forms/download/xef.html?filename=FPGAs_AdaptiveSoCs_Unified_SDI_2025.1_0530_0145_Win64.exe

- Peripheral features
 - USB-JTAG and USB-UART connection (FT2232H)
 - Power: DC-JACK 5V/1A
 - 16Mbit Serial flash memory
 - 16x LED
 - 1x 7-Segment (4 Digit FND)
 - 1x Buzzer
 - 6x DIP Switch
 - 5x Key, 5x Key LED
 - 1x Reset Key
 - 2x Sensor interface (I2C)

부품 배치도



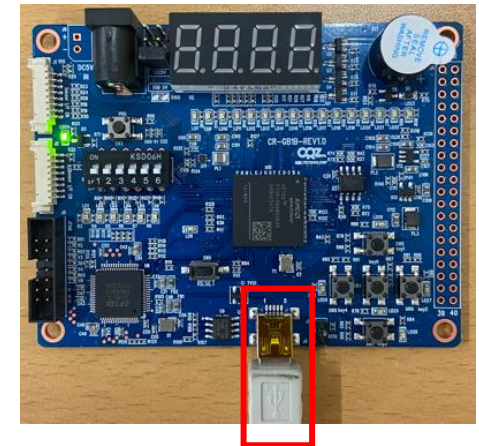
JTAG 연결

- USB 연결은 3가지 목적으로 사용
 - JTAG, UART, Power 공급
 - FT2232H for USB-JTAG and USB-UART connection
- 처음 연결 시 FT2232H Driver를 download해야 함
 - <http://www.ftdichip.com/Drivers/VCP.htm>

The screenshot shows the FTDI website with a red box highlighting the 'Currently Supported VCP Drivers' section. A red arrow points from the 'X64 (64-Bit)' link to the word '다운로드' (Download).

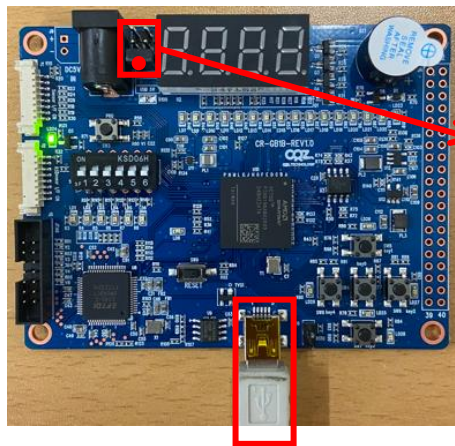
Operating System	Release Date	X86 (32-Bit)	X64 (64-Bit)
Windows (Desktop)*	2025-03-04	2.12.36.20	2.12.36.20

다운로드

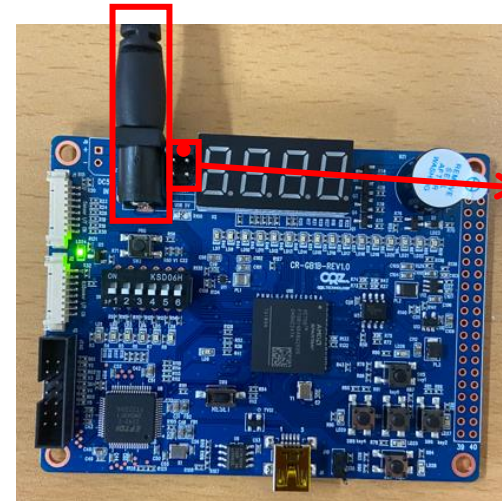


보드에 Power 공급하는 방법

- 보드에 power를 공급하는 방법은 2가지가 있으며, 사용하고자 하는 방법에 맞게 Jumper의 위치를 select해 주어야 함



1. USB 공급



2. Cable 공급

Jumper: 3번째 위치

Jumper: 1번째 위치

Reset Key를 이용한 reset 구현 (1/2)

- 설명

- Goldenbell1B 보드에는 Reset Key가 존재함
- Reset Key(pin M10)를 이용하여 program 내부에서 reset을 구현하여 사용

- Pin assignment

- Reset Key(pin M10)에 reset 신호를 연결

```
set_property IOSTANDARD LVCMOS33 [get_ports reset]
```

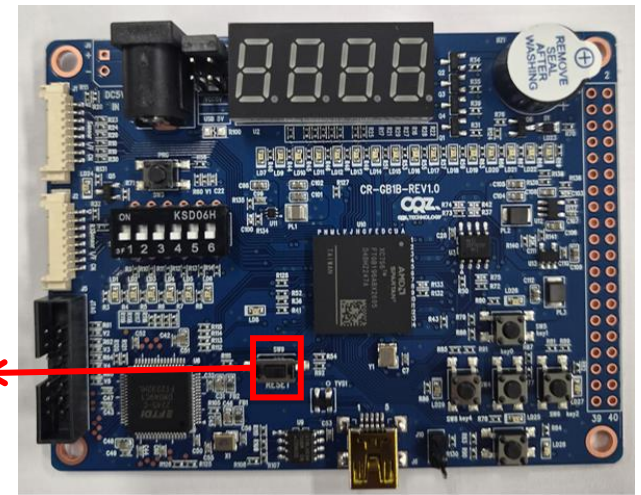
```
set_property PACKAGE_PIN M10 [get_ports reset]
```

Reset Key를 이용한 reset 구현 (2/2)

- 동작 방법
 - Reset: Reset Key를 누름
 - Reset 해제: Reset Key를 땀
- Source code

```
always @(posedge clk or negedge reset)
begin
    if(!reset)
    begin
        //이 자리에 reset시 동작 code 입력
    end
    else
    begin
        //이 자리에 reset이 아닐 시 동작 code 입력
    end
end
```

Reset Key ←



Peripheral Validation

LED 구동 (1/2)

- 방법

- LED에 1을 인가하면 on상태, 0을 인가하면 off상태가 됨

- Example 동작 방법

- Reset시, 모든 LED off
 - 모든 LED에 0을 인가
 - Reset이 아닐 시, led[7:4] & led[15:12] on
 - on 시킬 led (led[7:4] & led[15:12])에 1을 인가

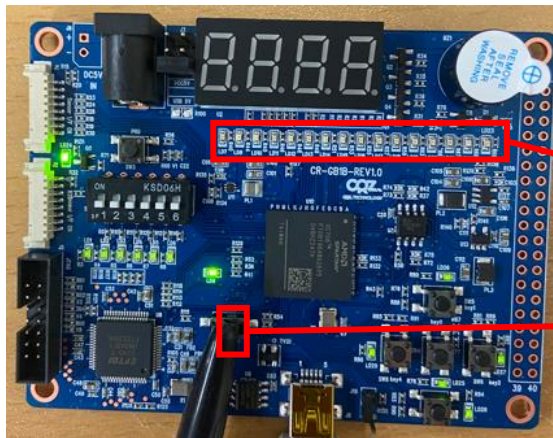
- Source code

- led_test.v

```
always @(*)
begin
  if(!reset)
    begin
      led <= 16'b0;
    end
  else
    begin
      led <= 16'b1111_0000_1111_0000;
    end
  end
end
```

LED 구동 (2/2)

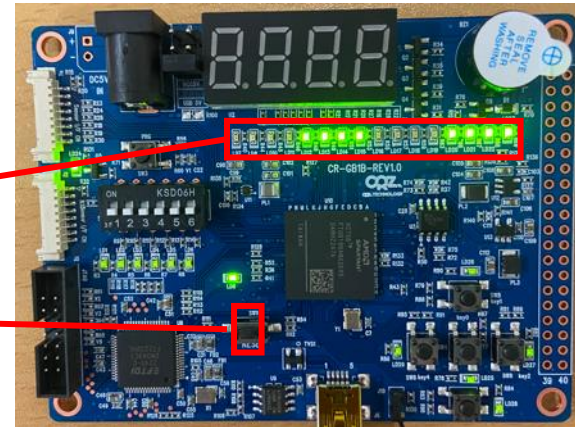
- Result
 - Reset 시, 모든 LED off
 - Reset 아닐 시, led[7:4] & led[15:12] on



Reset 상태

led[0]-led[15]

Reset 스위치



Non-reset 상태

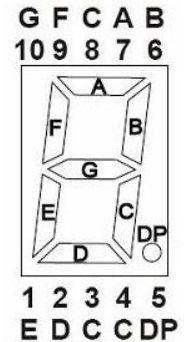
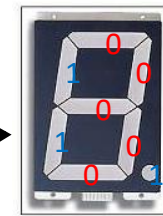
FND 구동 (1/3)

- FND 제어 방법

- FND는 0을 인가하면 on상태, 1을 인가하면 off상태

- 예) digit[3], digit[1]에 동시에 3을 출력하는 경우

```
digit<=4'b0101; fnd<=8'b00001101;
```



- 동작 방법

- Reset시, 모든 digit에 0 출력

- Reset이 아닐 시, digit[3]=3 digit[2]=2 digit[1]=1 digit[0]=0 출력

- 4개의 FND에 서로 다른 숫자 출력

- digit_sel은 제어할 digit을 선택해주는 register

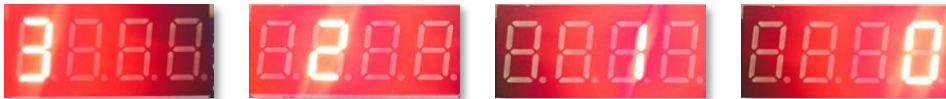
- Counter가 증가함에 따라서 digit_sel 값을 0,1,2,3 바뀌가며 각 digit_sel 값에 한 digit의 값을 출력 (시간 차를 두고 FND가 켜지는 효과 이용)

FND 구동 (2/3)

- Source code

- fnd_test.v

```
begin
    digit_sel <= digit_sel + 1'b1;
    case(digit_sel)
        0: begin digit <= 4'b1110; fnd <= 8'b00000011; end
        16384: begin digit <= 4'b1101; fnd <= 8'b10011111; end
        32768: begin digit <= 4'b1011; fnd <= 8'b00100101; end
        49152: begin digit <= 4'b0111; fnd <= 8'b00001101; end
    endcase
end
```

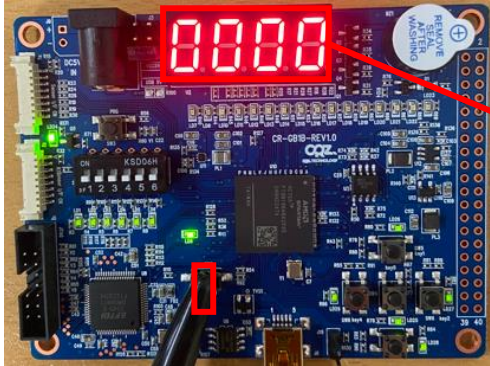


의 빠른 출력으로 →  이 동시에 출력되는 것처럼 보임

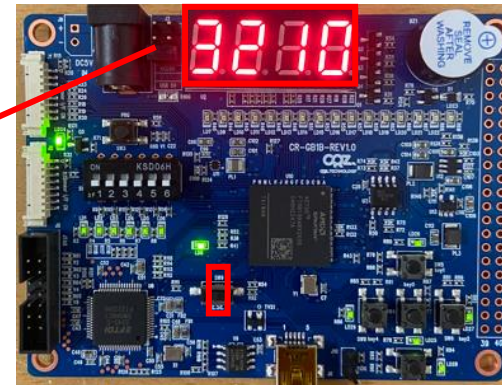
FND	a	b	c	d	e	f	g	.	2진수
0	0	0	0	0	0	0	1	1	8'b00000011
1	1	0	0	1	1	1	1	1	8'b10011111
2	0	0	1	0	0	1	0	1	8'b00100101
3	0	0	0	0	1	1	0	1	8'b00001101
4	1	0	0	1	1	0	0	1	8'b10011001
5	0	1	0	0	1	0	0	1	8'b01001001
6	0	1	0	0	0	0	0	1	8'b01000001
7	0	0	0	1	1	0	1	1	8'b00011011
8	0	0	0	0	0	0	0	1	8'b00000001
9	0	0	0	1	1	0	0	1	8'b00011001
A	1	0	0	0	1	0	0	1	8'b10001001
B	1	0	0	0	0	0	1	1	8'b10000011
C	0	1	1	0	0	0	1	1	8'b01100011
D	1	0	0	0	0	1	0	1	8'b10000101
E	0	1	1	0	0	0	0	1	8'b01100001
F	0	1	1	1	0	0	0	1	8'b01110001

FND 구동 (3/3)

- Result
 - Reset시, 모든 digit에 0 출력
 - Reset이 아닐 시, digit[3]=3 digit[2]=2 digit[1]=1 digit[0]=0 출력



Reset 상태



Non-reset 상태

digit[3]-digit[0]

Key & Key LED 구동 (1/2)

- Key 입력 제어
 - 눌렀을 때 1, Normal 0 인 상태
- Key LED 출력 제어
 - LED에 0을 인가하면 LED on, 1을 인가하면 LED off
- Example 동작 방법
 - Reset시, 모든 `keyLed<=5'b11111` 을 해주어 off 상태를 만들어 줌
 - Reset이 아닐 시, 각 key를 누르면 key의 상태가 1이므로 keyLed에 0을 인가해 주어 켜주기 위해 `~key`를 keyLed에 대입

Key & Key LED 구동 (2/2)

- Source code
 - Key_test.v

```
begin
  if(~reset)
    begin
      KeyLed <= 5'b11111;
    end
  else
    begin
      KeyLed <= ~key;
    end
  end
end
```

- Result
 - Reset시, 모든 LED off
 - Reset이 아닐 시, Key를 push했을 때 각각의 key에 해당하는 key LED on



key[1]
key[4] key[0] key[2]
key[3]

Dip Switch 구동 (1/2)

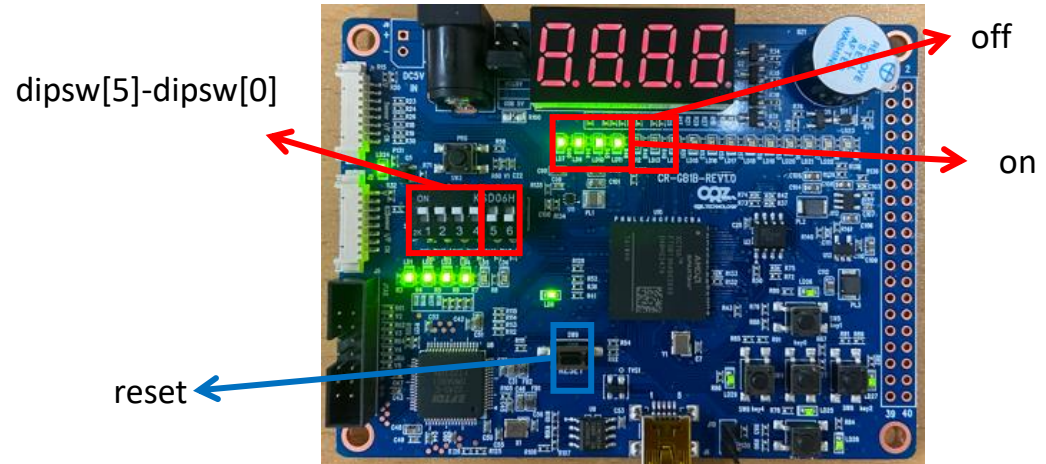
- Dip switch 제어 방법
 - On일 때 0, off일 때 1
- 동작 방법
 - dip switch[5:0]로 led[5:0]를 제어
 - Reset시, 모든 LED off 상태
 - Reset이 아닐 시, dip switch off → LED off & dip switch on → LED on
 - Dip switch는 off일 때, 1인 상태
 - Led는 1일 때 on되기 때문에, dip switch가 on(0)일 때 켜지게 하기 위해 ~dip switch를 LED에 대응 시킴
 - 즉, dip switch는 on일 때 0인 상태이지만 LED에는 ~dip switch가 대응 되어 있기 때문에 1이 인가됨 → LED on

Dip Switch 구동 (2/2)

- Source code
 - dip_switch.v

```
begin
  if(~reset)
    begin
      rLed <= 6'b000000;
    end
  else
    begin
      rLed <= ~dipsw;
    end
  end
end
```

- Result
 - Reset시, 모든 LED off 상태
 - Reset이 아닐 시, dip switch off → LED off
dip switch on → LED on
 - 상위 2-bit dip switch만 off 상태이므로 상위 2-bit LED off

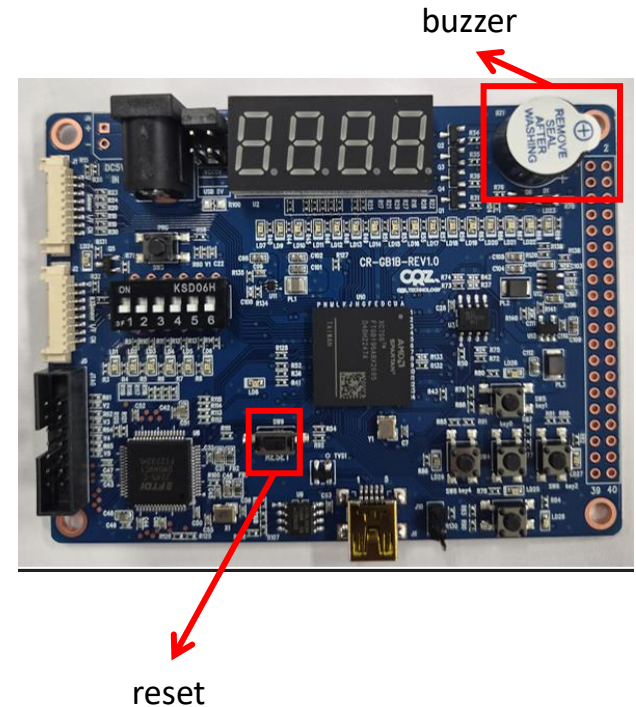


Buzzer 구동

- Buzzer 제어
 - 소리 On: 1
 - 소리 Off: 0
- 동작 방법
 - Reset으로 buzzer의 on, off 제어
- Source code
 - buzzer_test.v

```
if(~reset)
    rBuzzer <= 0;
else
    rBuzzer <= 1;
```

- Result
 - Reset시 buzz<=0 이므로, buzzer off
 - Program시 buzz<=1 이므로, buzzer on



Buzzer 구동 사례

- Example 동작방법

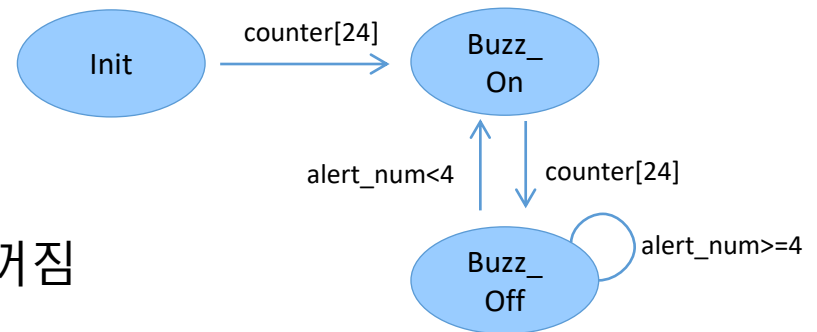
- Program시 buzzer가 1초 간격으로 4번 울리도록 함
- Buzz_On state에서 alert_num변수를 두고 울릴 때마다 count
- alert_num 4일 때 까지는 On,Off state를 반복하다 alert_num>4일 때 부터는 Buzz_Off state 유지

- Source code

- Buzzzer_ad_test

- Result

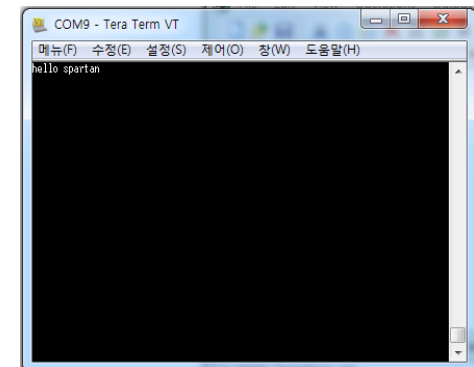
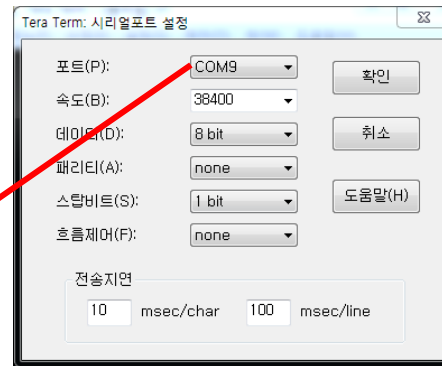
- buzzer가 1초 간격으로 4번 울렸다 꺼짐



UART Example

- Example 동작방법
 - Uart의 출력을 확인하려면 virtual terminal상에서 확인해야 함
 - Virtual terminal로 입력한 값(키보드입력) 그대로 출력
 - Virtual terminal software tool - Tera Term
- Source code
 - uart.v
- Result
 - 입력 값 그대로 출력

포트는 장치관리자에서 연결 포트를 확인 한 후 입력



Vivado Project 생성 방법

LED project를 생성하고 이를 FPGA에 프로그램

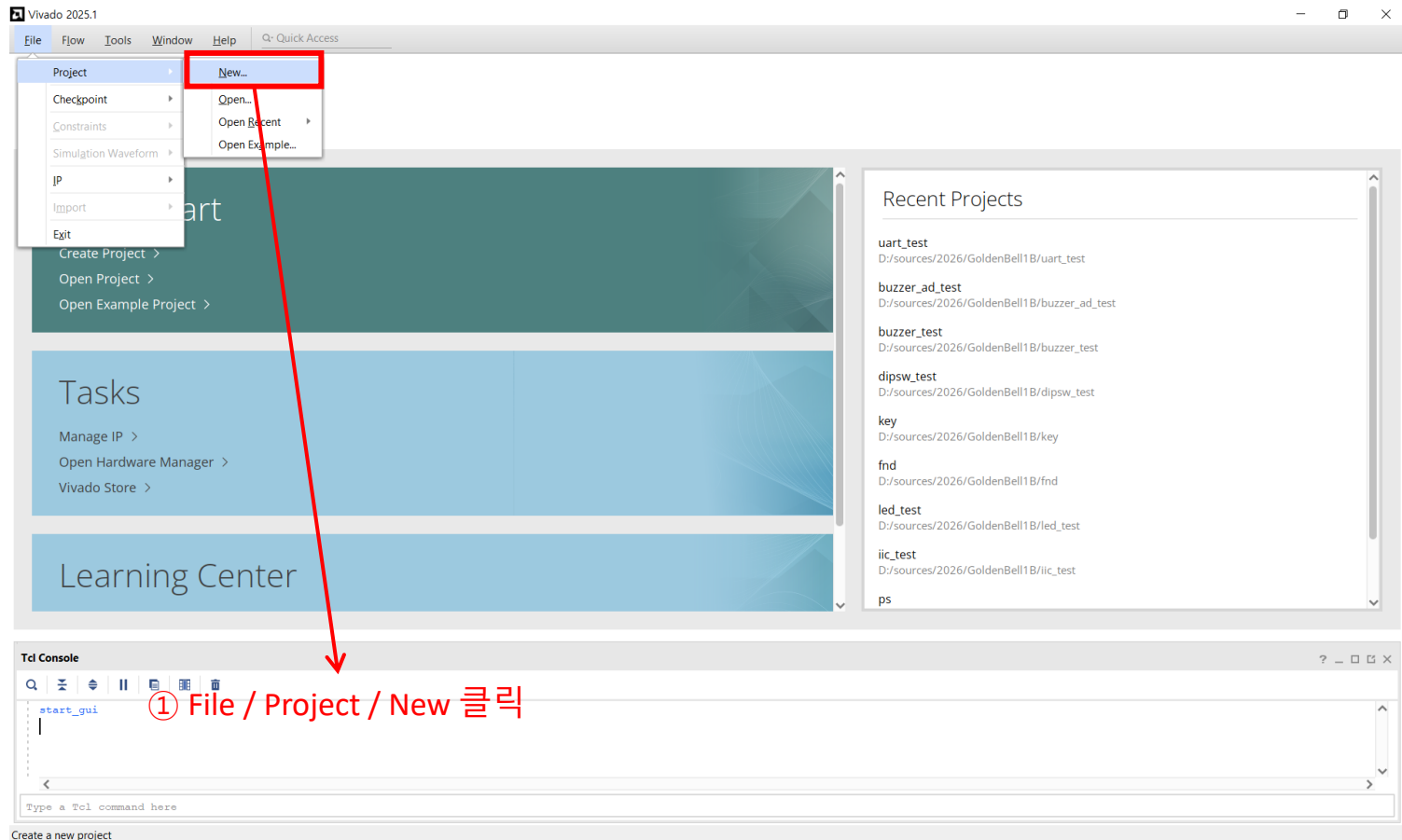
Step 1. Creating a new project

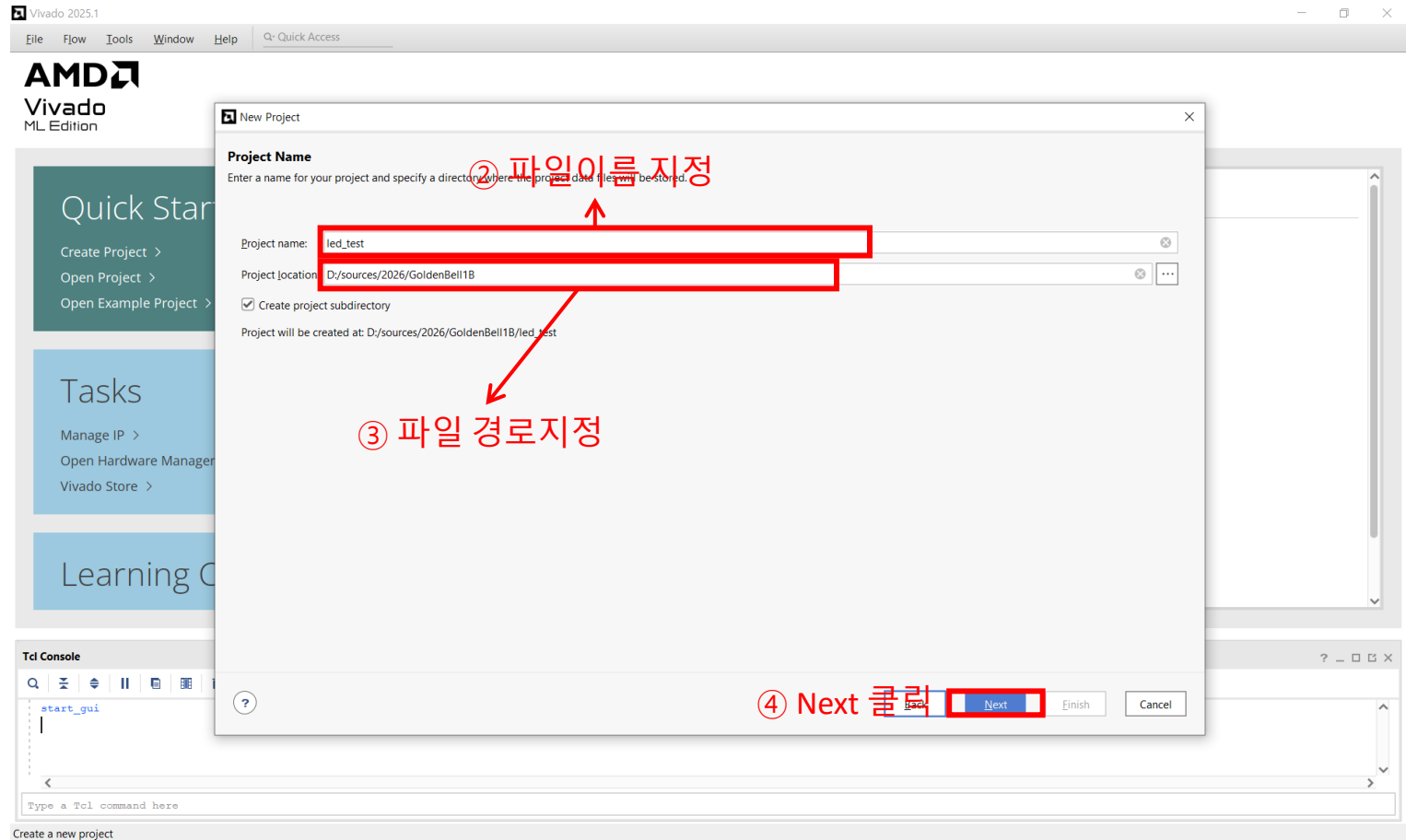
Step 2. Generating XDC file

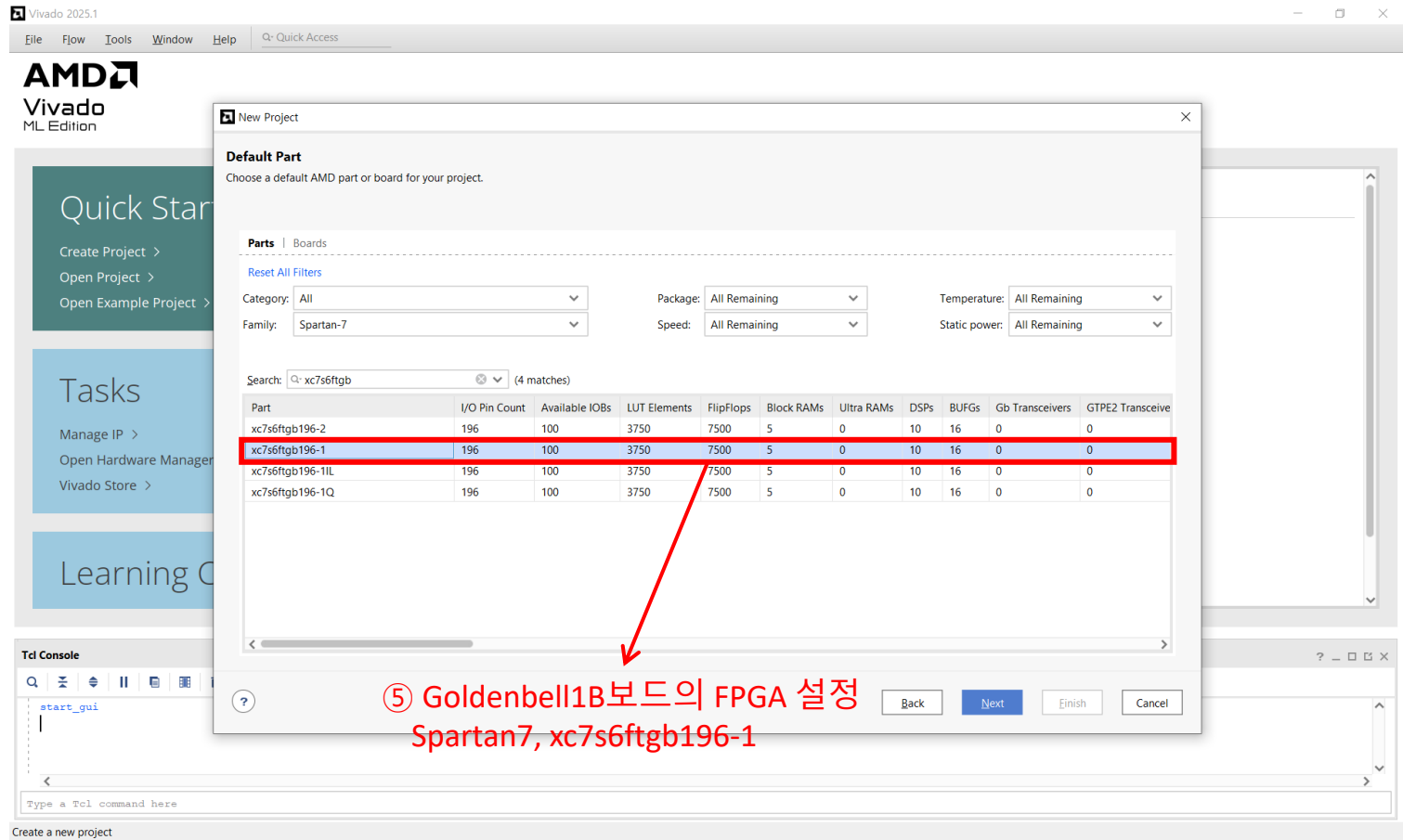
Step 3. Creating bit file

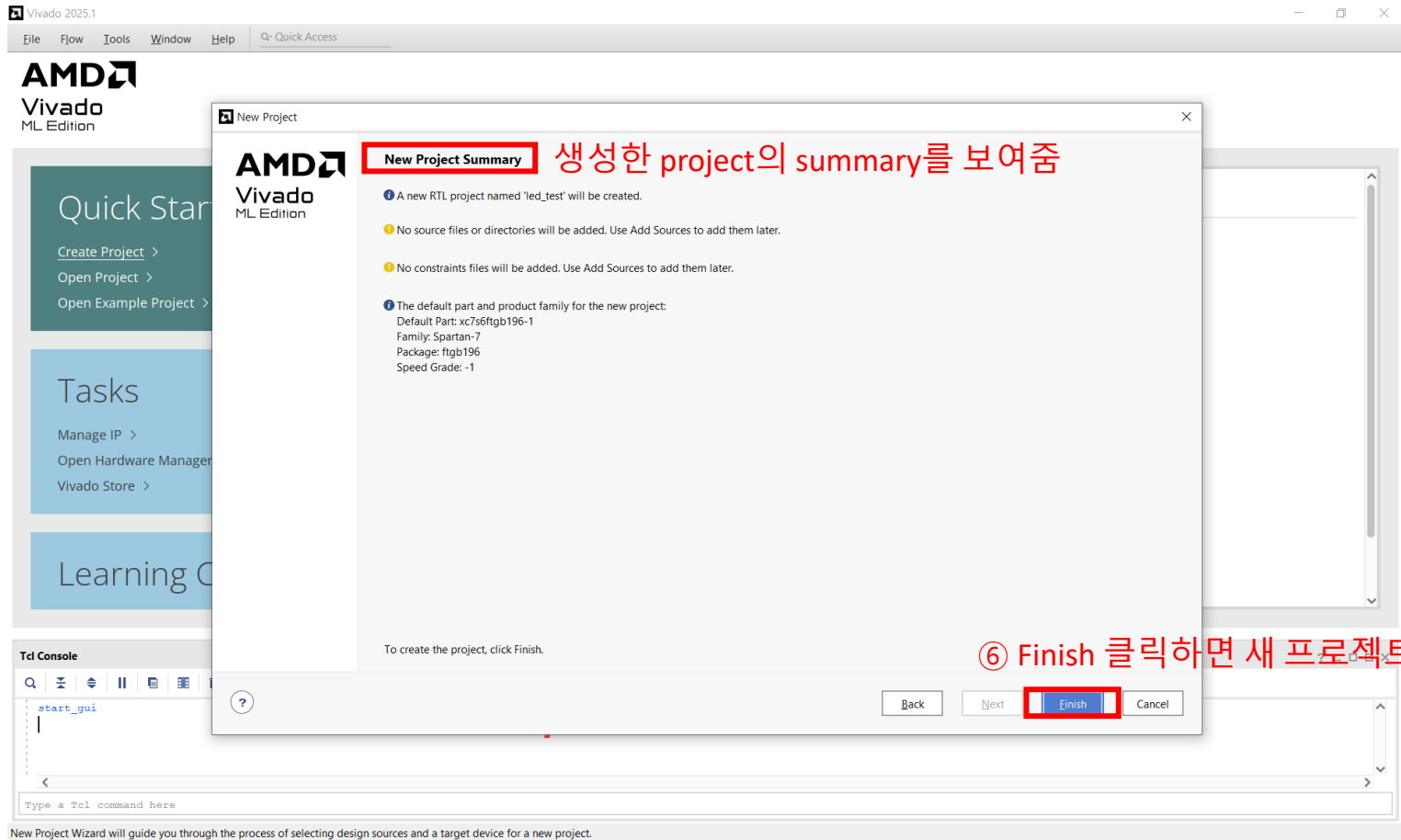
Step 4. Downloading a bit file to FPGA

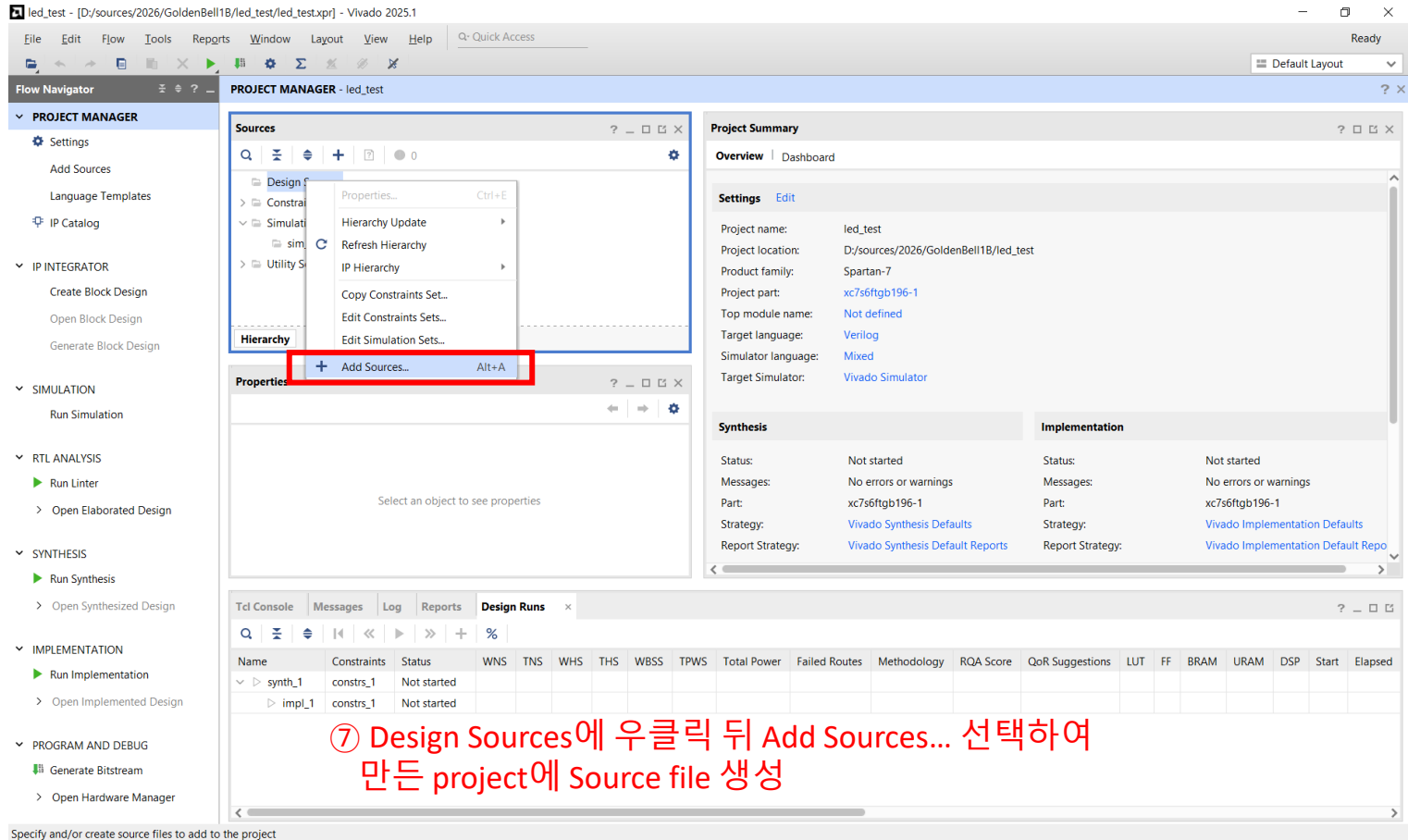
Step 1. Creating new project

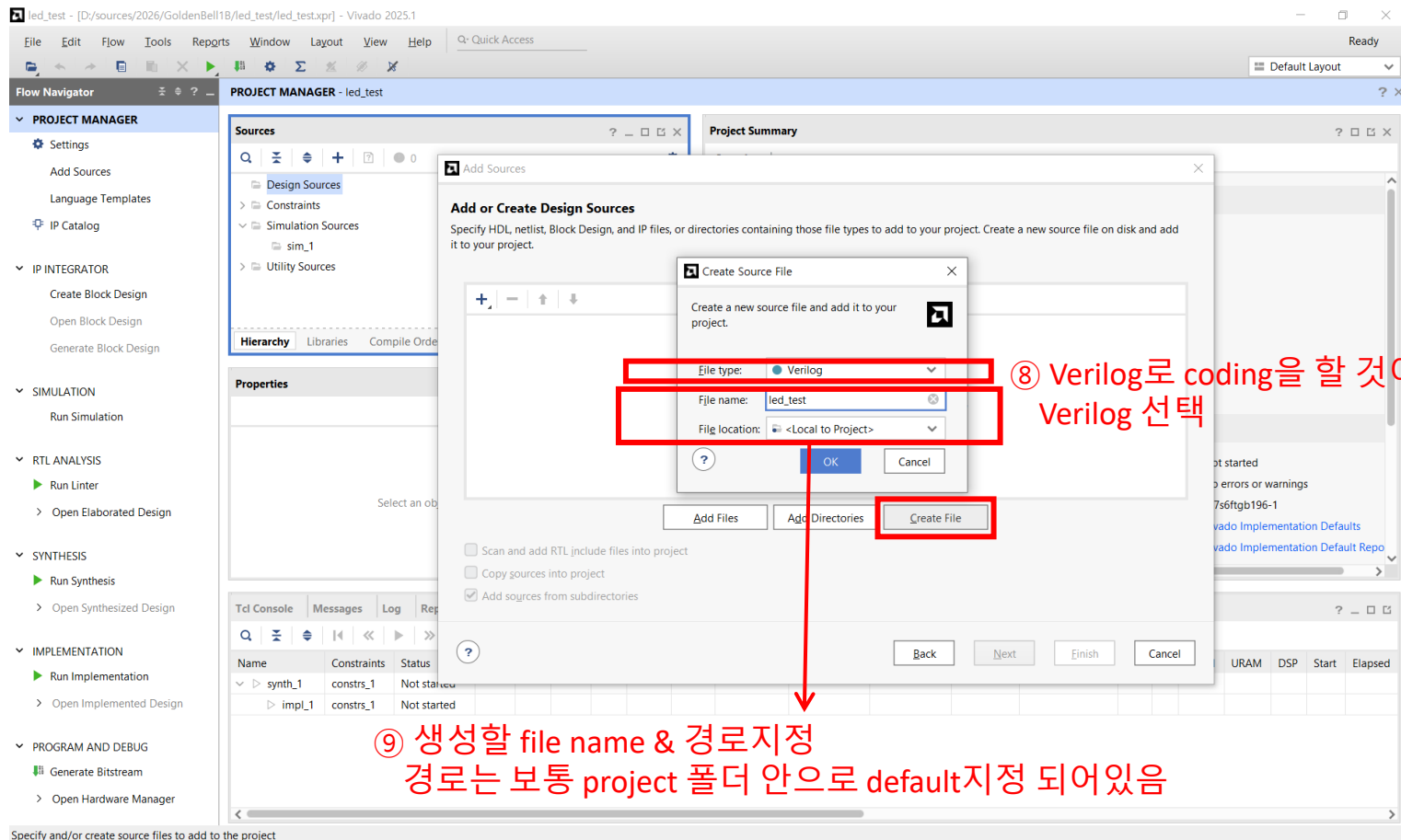


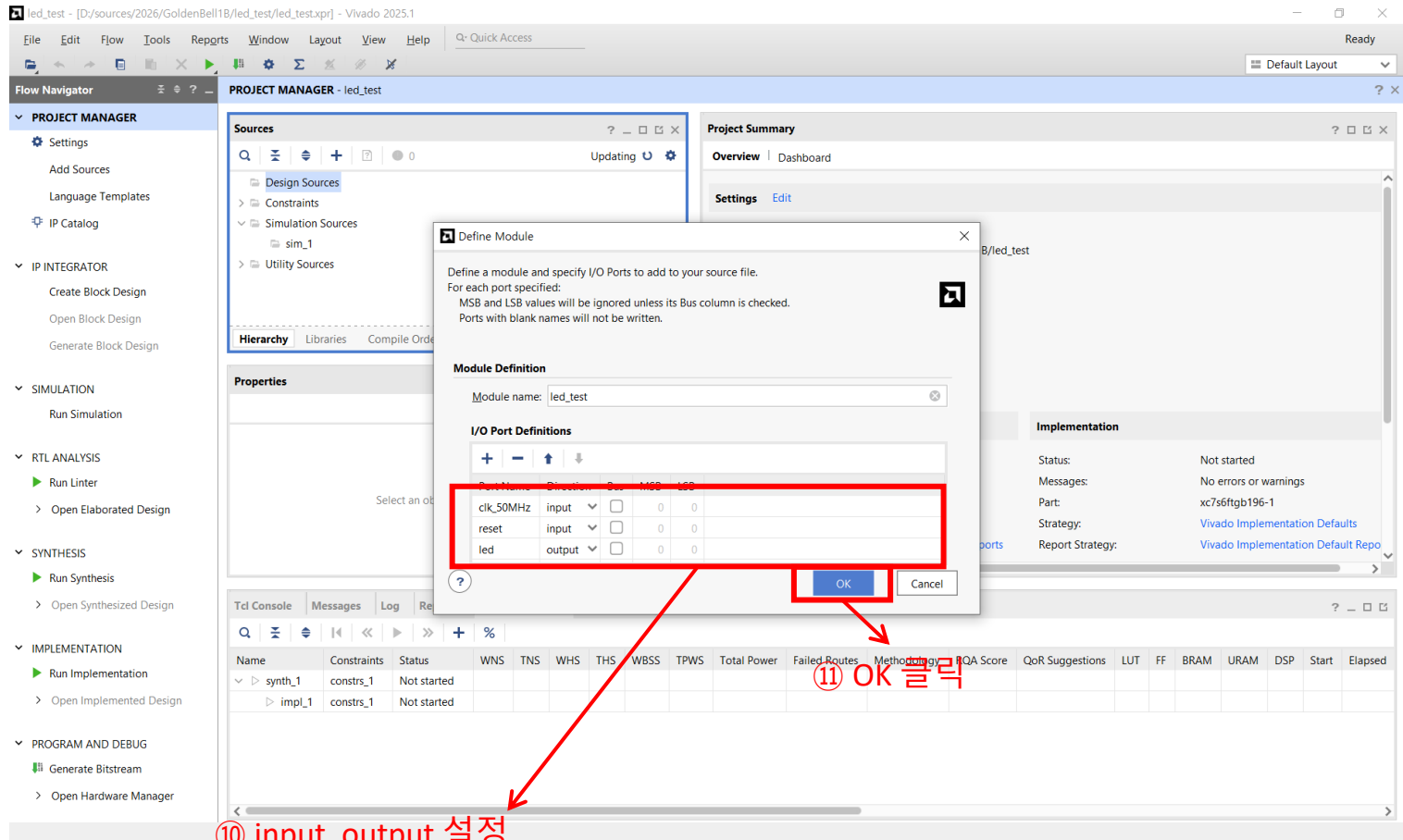


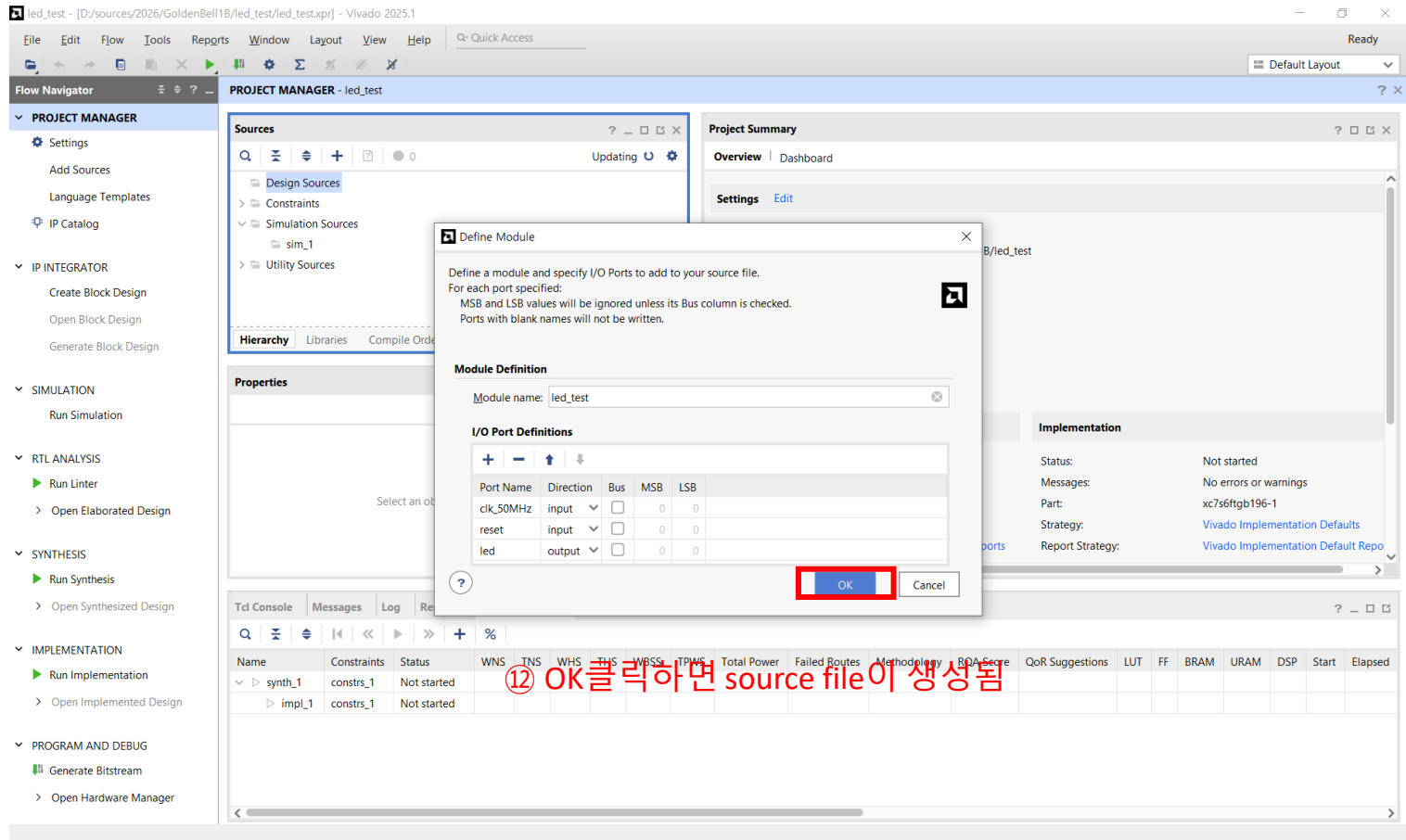












led_test - [D:/sources/2026/GoldenBell1B/led_test/led_test.xpr] - Vivado 2025.1

File Edit Flow Tools Reports Window Layout View Help Q: Quick Access

Ready

Default Layout

Flow Navigator PROJECT MANAGER - led_test

PROJECT MANAGER

- Settings
- Add Sources
- Language Templates
- IP Catalog
- IP INTEGRATOR
 - Create Block Design
 - Open Block Design
 - Generate Block Design
- SIMULATION
 - Run Simulation
- RTL ANALYSIS
 - Run Linter
 - Open Elaborated Design
- SYNTHESIS
 - Run Synthesis
 - Open Synthesized Design
- IMPLEMENTATION
 - Run Implementation
 - Open Implemented Design
- PROGRAM AND DEBUG
 - Generate Bitstream
 - Open Hardware Manager

Sources

Design Sources (1)

- led_test (led_test.v)

Simulation Sources (1)

- sim_1 (1)

Utility Sources

Hierarchy Libraries Compile Order

Source File Properties

led_test.v

Enabled

Location: D:/sources/2026/GoldenBell1B/led_test/led_test.srscs/sources_1/new

Type: Verilog

Library: xil_defaultlib

General Properties

Project Summary x led_test.v x

D:/sources/2026/GoldenBell1B/led_test/led_test.srscs/sources_1/new/led_test.v

```

9 // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //////////////////////////////////////////////////
21
22
23 module led_test(
24     input clk_50MHz,
25     input reset,
26     output led
27 );
28 endmodule
29

```

Source file 생성 끝 !!!

Tcl Console Messages Log Reports Design Runs x

Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodology	RQA Score	QoR Suggestions	LUT	FF	BRAM	URAM	DSP	Start	Elapsed
synth_1	constrs_1	Not started																		
impl_1	constrs_1	Not started																		

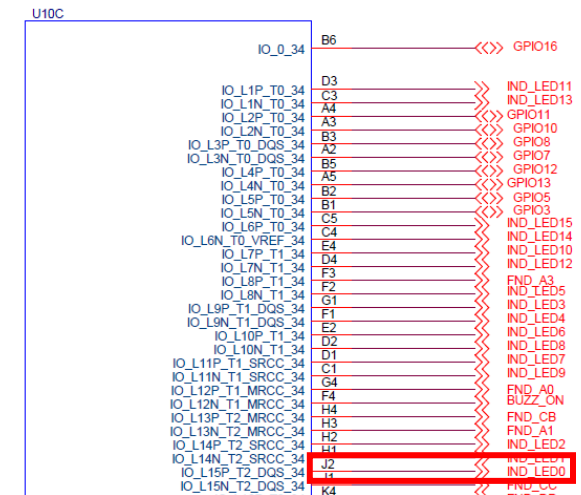
23:1 Insert Verilog

Step 2. Generating XDC-file

Pin map을 위한 XDC파일 생성 방법

XDC File Generation Example

- example.xdc 파일을 프로젝트에 추가한다
 - example.xdc 파일은 peripheral_test_code 폴더에 존재
 - 현재 working directory에 .xdc file을 옮김
- 보드의 회로도를 확인하고 원하는 input과 output의 번호를 확인
 - 예) Output으로 선언한 led_output으로 Led0[J2 pin]를 제어하고 싶을 경우
 - Led0가 J2에 mapping되어있음을 확인
 - XDC 파일에 아래 문법에 맞춰 입력
 - 문법: `set_property PACKAGE_PIN "pin#" [get_ports {"output_name"}];`
 - ex) `set_property PACKAGE_PIN J2 [get_ports {led_output}];`



led_test - [D:/sources/2026/GoldenBell1B/led_test/led_test.xpr] - Vivado 2025.1

File Edit Flow Tools Reports Window Layout View Help Q: Quick Access

Synthesis and Implementation Out-of-date details

Default Layout

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Sources

① Constraints 우 클릭

Design Sources (1)

- led_test (led_test)
- Constr...
- Simul...
- Utility

Hierarchy

Source File

led_test.d + Add Sources... Alt+A

Enabled

Location: D:/sources/2026/GoldenBell1B/led_test/srcs/utlis_1/imports/syn

Type: Design Checkpoint

Size: 11.0 KB

General Properties

Project Summary

Overview Dashboard

Settings Edit

Project name: led_test

Project location: D:/sources/2026/GoldenBell1B/led_test

Product family: Spartan-7

Project part: xc7s6ftgb196-1

Top module name: led_test

Target language: Verilog

Simulator language: Mixed

Target Simulator: Vivado Simulator

Synthesis Implementation Summary Route Status

Status: Out-of-date

Messages: 10 warnings

Part: xc7s6ftgb196-1

Strategy: Vivado Synthesis Defaults

Report Strategy: Vivado Synthesis Default Reports

Status: Out-of-date

Messages: 2 warnings

Part: xc7s6ftgb196-1

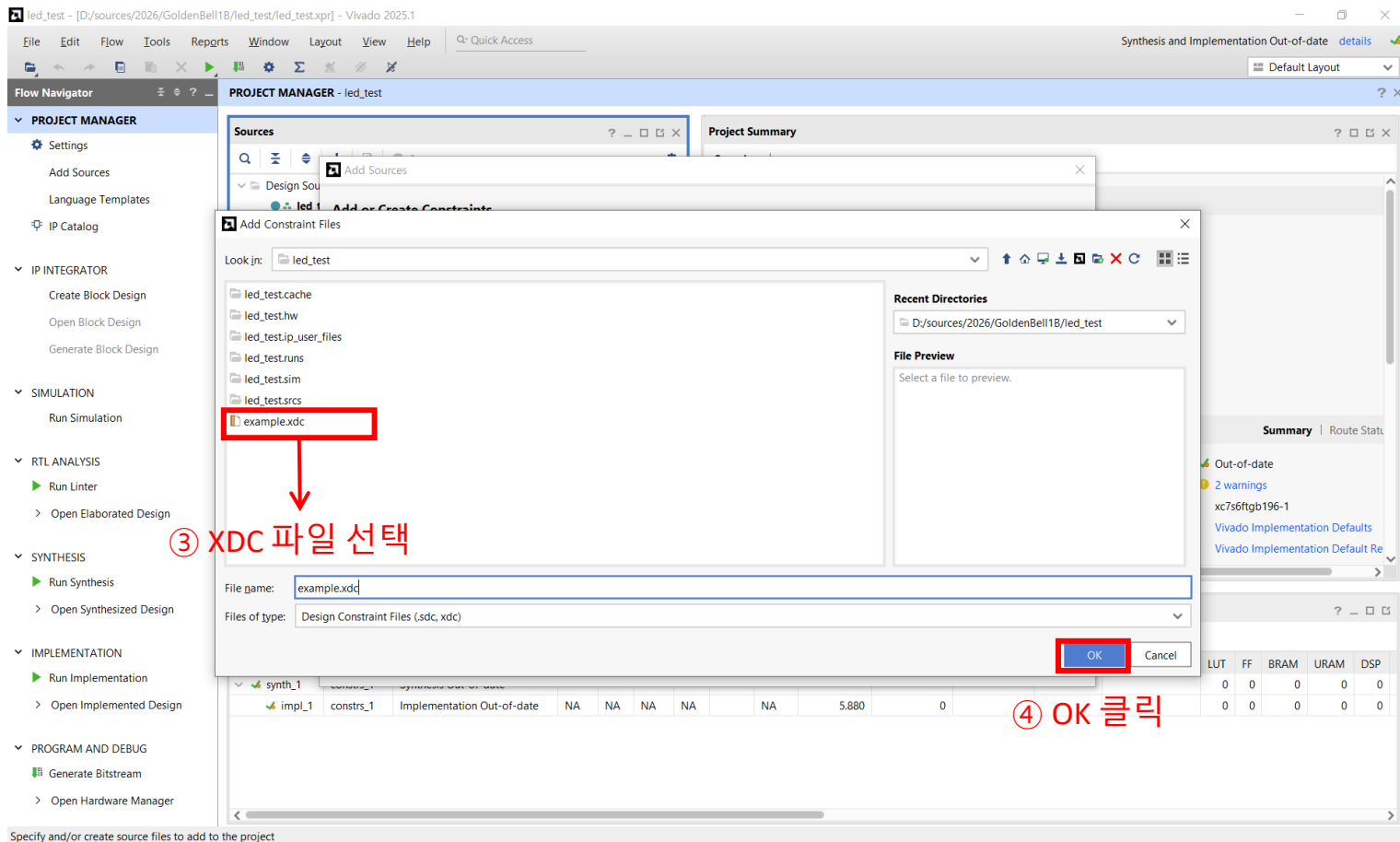
Strategy: Vivado Implementation Defaults

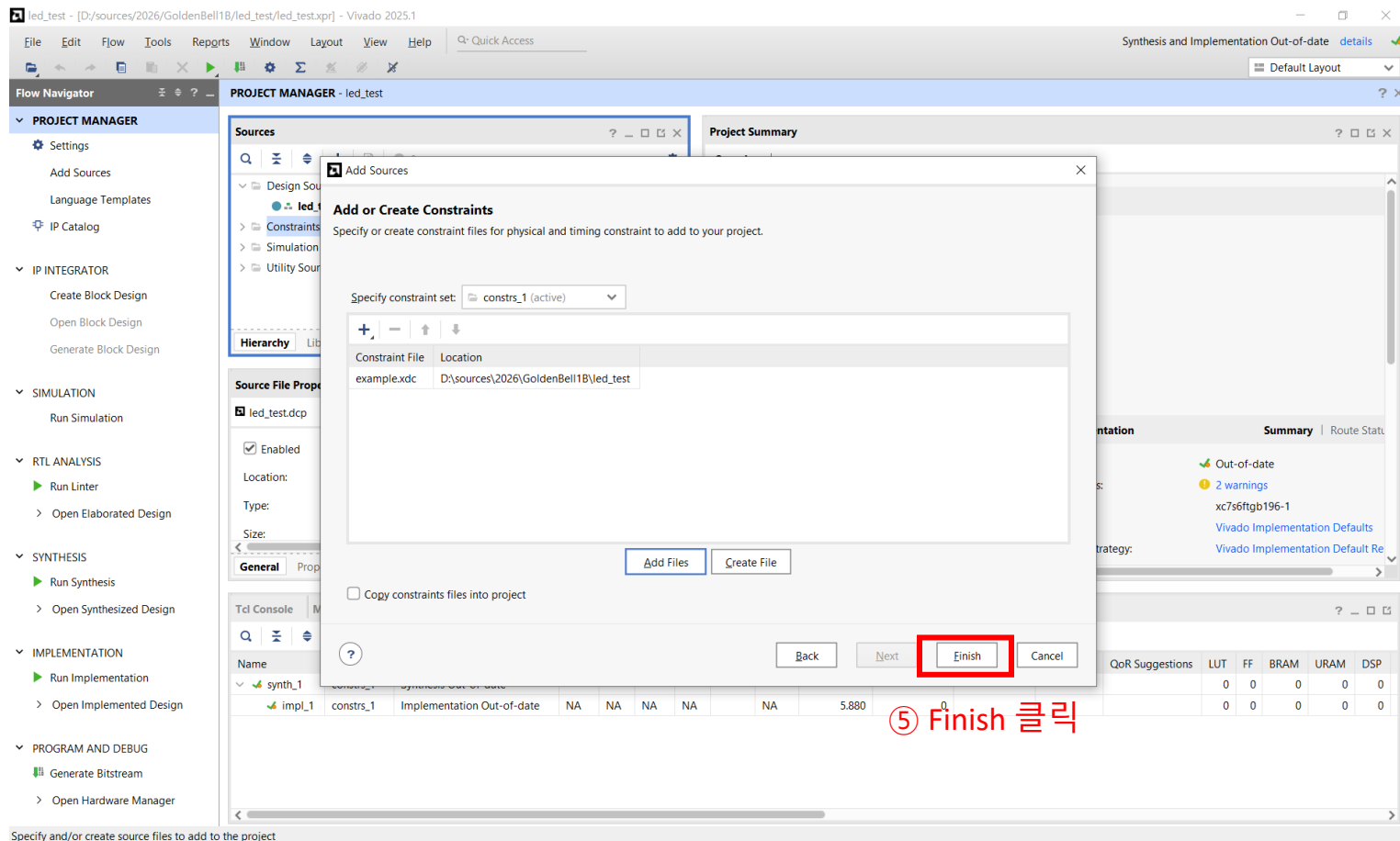
Report Strategy: Vivado Implementation Default Reports

Tcl Console Messages Log Reports Design Runs

Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodology	RQA Score	QoR Suggestions	LUT	FF	BRAM	URAM	DSP
synth_1	constrs_1	Synthesis Out-of-date												0	0	0	0	0
impl_1	constrs_1	Implementation Out-of-date	NA	NA	NA	NA		NA	5.880	0				0	0	0	0	0

Specify and/or create source files to add to the project





led_test - [D:/sources/2026/GoldenBell1B/led_test/led_test.xpr] - Vivado 2025.1

File Edit Flow Tools Reports Window Layout View Help Q: Quick Access Synthesis and Implementation Out-of-date details Default Layout

Flow Navigator PROJECT MANAGER - led_test

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- SYNTHESIS
 - Run Synthesis
 - Open Synthesized Design
- IMPLEMENTATION
 - Run Implementation
 - Open Implemented Design
- PROGRAM AND DEBUG
 - Generate Bitstream
 - Open Hardware Manager

Sources

Design Sources (1)

- led_test (led_test.v)

Constraints (1)

- constrs_1 (1)
 - example.xdc

Simulation Sources (1)

- Utility Sources (1)

Hierarchy Libraries Compile Order

Source File Properties

example.xdc

Enabled

Location: D:/sources/2026/GoldenBell1B/led_test

Type: XDC

Size: 1.8 KB

General Properties

Project Summary led_test.v

D:/sources/2026/GoldenBell1B/led_test/srcs/sources_1/new/led_test.v

```

13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module led_test(
24     reset,
25     led
26 )
27
28     input reset;
29     output reg [15:0] led;
30
31     always @(*)
32     begin
33         if (!reset)

```

example.xdc 파일 프로젝트 안에 넣기 끝!!!
이제....xdc 파일을 수정하여 쓰시면 됩니다

Tcl Console Messages Log Reports Design Runs

Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodology	RQA Score	QoR Suggestions	LUT	FF	BRAM	URAM	DSP
synth_1	constrs_1	Synthesis Out-of-date												0	0	0	0	0
impl_1	constrs_1	Implementation Out-of-date	NA	NA	NA	NA		NA	5.880	0				0	0	0	0	0

Redo the last undone edit

Step 3. Creating a Bit File

보드에 programming하기 위한 bit-file을 생성하는
방법

led_test - [D:/sources/2026/GoldenBell1B/led_test/led_test.xpr] - Vivado 2025.1

File Edit Flow Tools Reports Window Layout View Help Q: Quick Access

Synthesis and Implementation Out-of-date details

Default Layout

Flow Navigator PROJECT MANAGER - led_test

PROJECT MANAGER

- Settings
- Add Sources
- Language Templates
- IP Catalog

IP INTEGRATOR

- Create Block Design
- Open Block Design
- Generate Block Design

SIMULATION

- Run Simulation

RTL ANALYSIS

- Run Linter
- Open Elaborated Design

SYNTHESIS

- Run Synthesis
- Open Synthesized Design

IMPLEMENTATION

- Run Implementation
- Open Implemented Design

PROGRAM AND DEBUG

- Generate Bitstream**
- Open Hardware Manager

PROJECT MANAGER - led_test

Sources

- Design Sources (1)
 - led_test (led_test.v)
- Constraints (1)
 - constrs_1 (1)
 - example.xdc
- Simulation Sources (1)
- Utility Sources (1)

Hierarchy Libraries Compile Order

Source File Properties

example.xdc

Enabled

Location: D:/sources/2026/GoldenBell1B/led_test

Type: XDC

Size: 10.4 KB

General Properties

Tcl Console

Project Summary

led_test.v

D:/sources/2026/GoldenBell1B/led_test/srcs/sources_1/new/led_test.v

```

13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module led_test(
24     reset,
25     led
26 );
27
28 input reset;
29 output reg [15:0] led;
30
31 //
32 //
33 //
34 //
35 //
36 //
37 //
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40 //
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85 //
86 //
87 //
88 //
89 //
90 //
91 //
92 //
93 //
94 //
95 //
96 //
97 //
98 //
99 //
100 //

```

Generate Bitstream

Generate a programming file after implementation

① .xdc 파일까지 수정을 완료 하였으면, bit파일을 생성함
Generate Bitstream을 클릭하면 bit파일이 생성됨
Synthesize -> Implement design -> Generate Bitstream 순서로 진행

Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodology	RQA Score	QoR Suggestions	LUT	FF	BRAM	URAM	DSP
synth_1	constrs_1	Synthesis Out-of-date												0	0	0	0	0
impl_1	constrs_1	Implementation Out-of-date	NA	NA	NA	NA		NA	5.880	0				0	0	0	0	0

FPGA programming

- FPGA board에 programming하는 방법 2가지 존재
 - 1. bit file FPGA downloading
 - .bit file을 직접 FPGA에 다운로드 하여 바로 실행
 - 2. SPI Flash programming
 - SPI Flash에 programming하여 booting시(전원 인가 시) 바로 실행
 - 이를 위해 먼저 .bin file을 생성해야 함

Step 4. Downloading a Bit File to FPGA

생성한 bit-file을 보드에 download하는 방법

led_test - [D:/sources/2026/GoldenBell1B/led_test/led_test.xpr] - Vivado 2025.1

File Edit Flow Tools Repgrts Window Layout View Help Q: Quick Access write_bitstream Complete ✓ Default Layout

Flow Navigator PROJECT MANAGER - led_test

PROJECT MANAGER

- Settings
- Add Sources
- Language Templates
- IP Catalog

IP INTEGRATOR

- Create Block Design
- Open Block Design
- Generate Block Design

SIMULATION

- Run Simulation

RTL ANALYSIS

- Run Linter
- Open Elaborated Design

SYNTHESIS

- Run Synthesis
- Open Synthesized Design

IMPLEMENTATION

- Run Implementation
- Open Implemented Design

PROGRAM AND DEBUG

- Generate Bitstream
- Open Hardware Manager

Open the hardware program and debug manager

Project Summary led_test.v

D:/sources/2026/GoldenBell1B/led_test/led_test.srcs/sources_1/new/led_test.v

```

24     reset,
25     led
26 );
27
28 input reset;
29 output reg [15:0] led;
30
31 always @(*)
32 begin
33 if(!reset)
34     begin
35         led <= 16'b0;
36     end
37 begin
38     led <= 16'b1111_0000_1111_0000;
39 end
40 end
41 end
42 endmodule
43
44

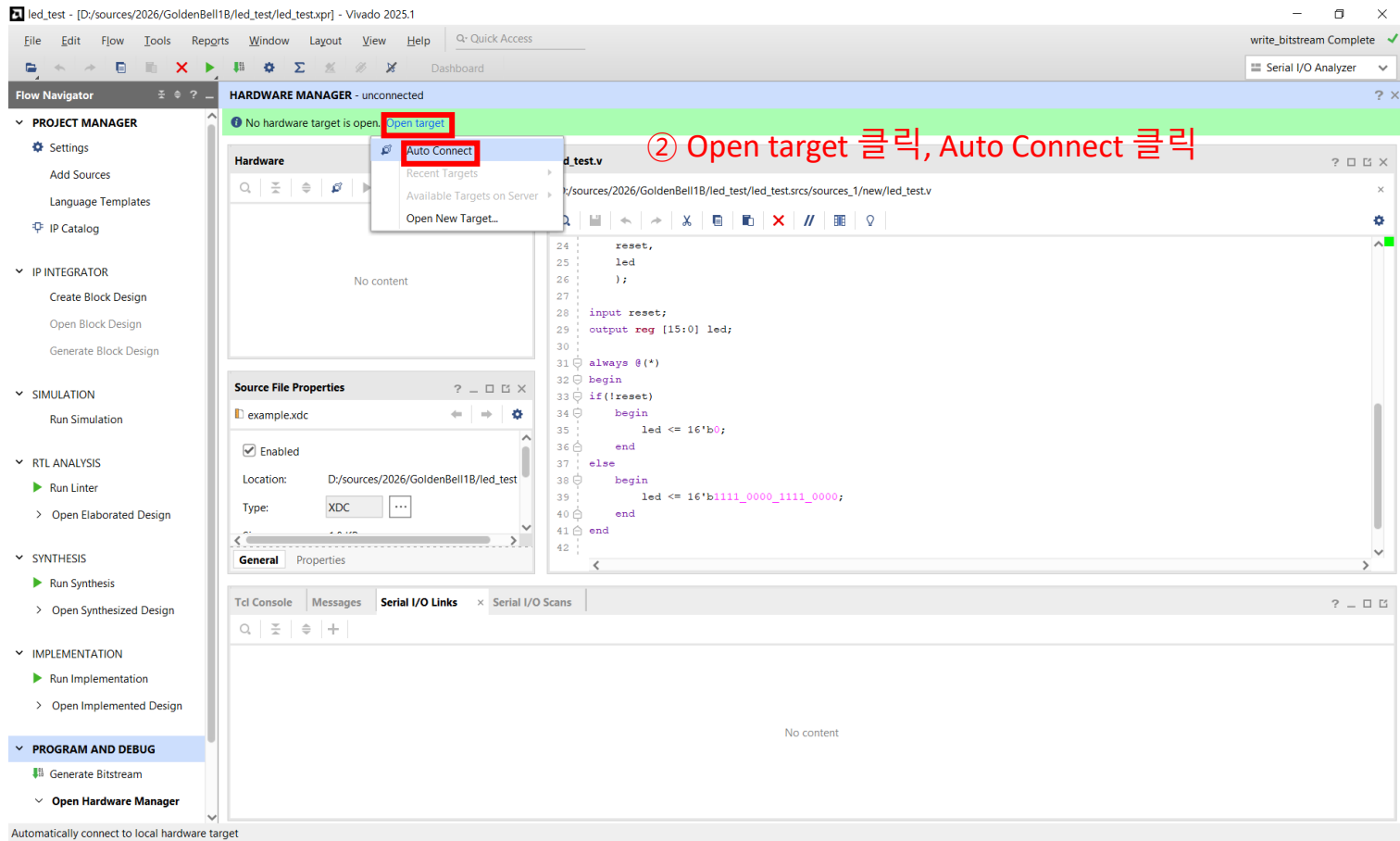
```

*이 flow는 bit file을 생성 했음을 가정으로 함

Tcl Console Messages Log Reports Design Runs

Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodology	RQA Score	QoR Suggestions	LUT	FF	BRAM	URAM	DSP
✓ synth_1	constrs_1	synth_design Complete!												0	0	0	0	0
✓ impl_1	constrs_1	write_bitstream Complete!	NA	NA	NA	NA		NA	5.880	0				0	0	0	0	0

① Open Hardware Manager를 클릭



led_test - [D:/sources/2026/GoldenBell1B/led_test/led_test.xpr] - Vivado 2025.1

File Edit Flow Tools Repgrts Window Layout View Help Q: Quick Access

write_bitstream Complete ✓

Serial I/O Analyzer

Flow Navigator

PROJECT MANAGER

- Settings
- Add Sources
- Language Templates
- IP Catalog

IP INTEGRATOR

- Create Block Design
- Open Block Design
- Generate Block Design

SIMULATION

- Run Simulation

RTL ANALYSIS

- Run Linter
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SYNTHESIS

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IMPLEMENTATION

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PROGRAM AND DEBUG

- Generate Bitstream
- Open Hardware Manager

HARDWARE MANAGER - localhost/xilinx_tcf/Digilent/210205329746A

There are no debug cores. Program device Refresh device

Hardware

Name	Status
localhost (1)	Connected
xilinx_tcf/Digilent/210205329746A	Open
xc7s6_0 (0)	Programmed

Hardware Target Properties

Name: localhost/xilinx_tcf/Digilent/210205329746A

Type: xilinx_tcf

Status: Open

led_test.v

```

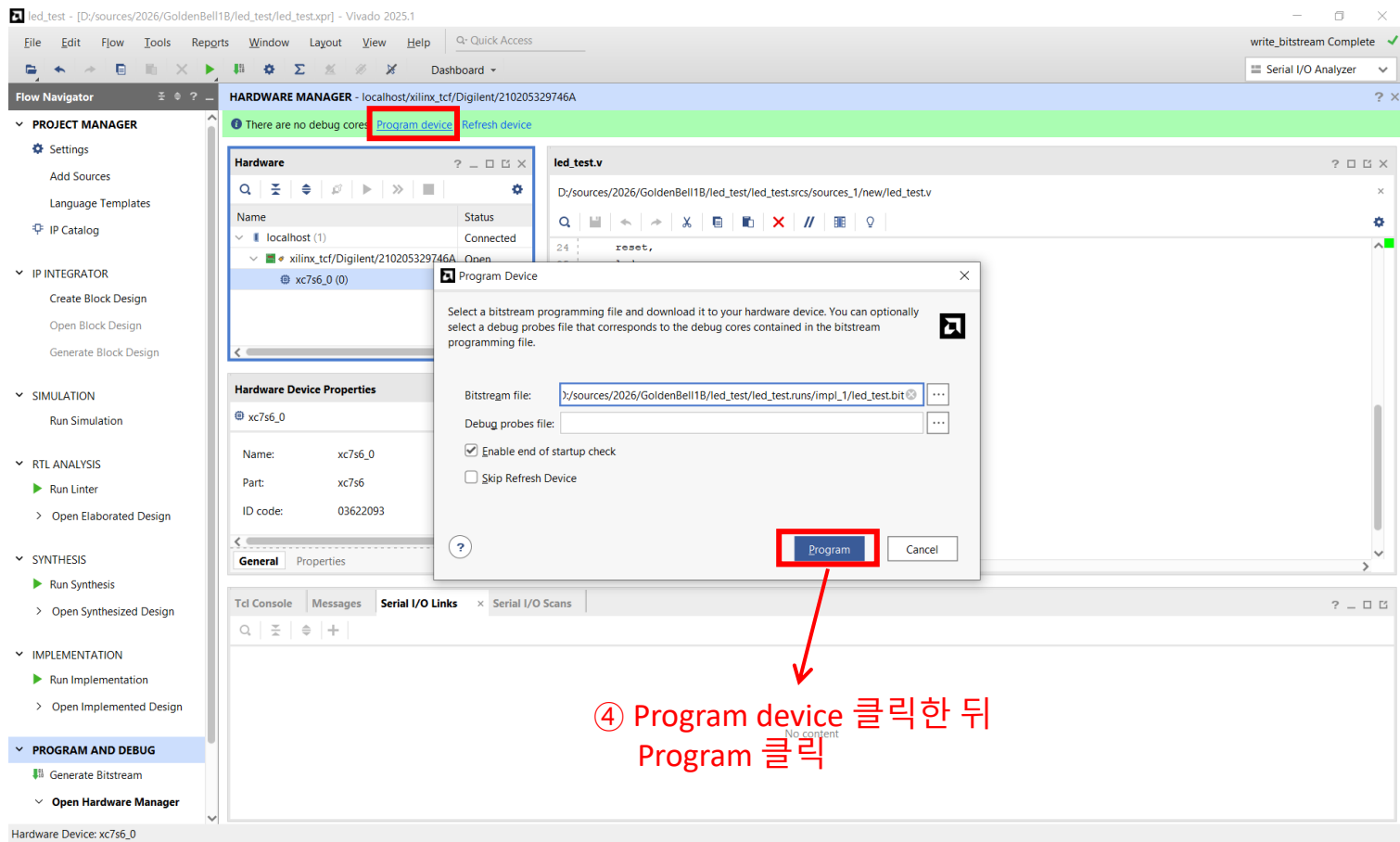
D:/sources/2026/GoldenBell1B/led_test/srcs/sources_1/new/led_test.v
24 reset,
25 led
26 );
27
28 input reset;
29 output reg [15:0] led;
30
31 always @(*)
32 begin
33 if(!reset)
34 begin
35 led <= 16'b0;
36 end
37 else
38 begin
39 led <= 16'b1111_0000_1111_0000;
40 end
41 end
42

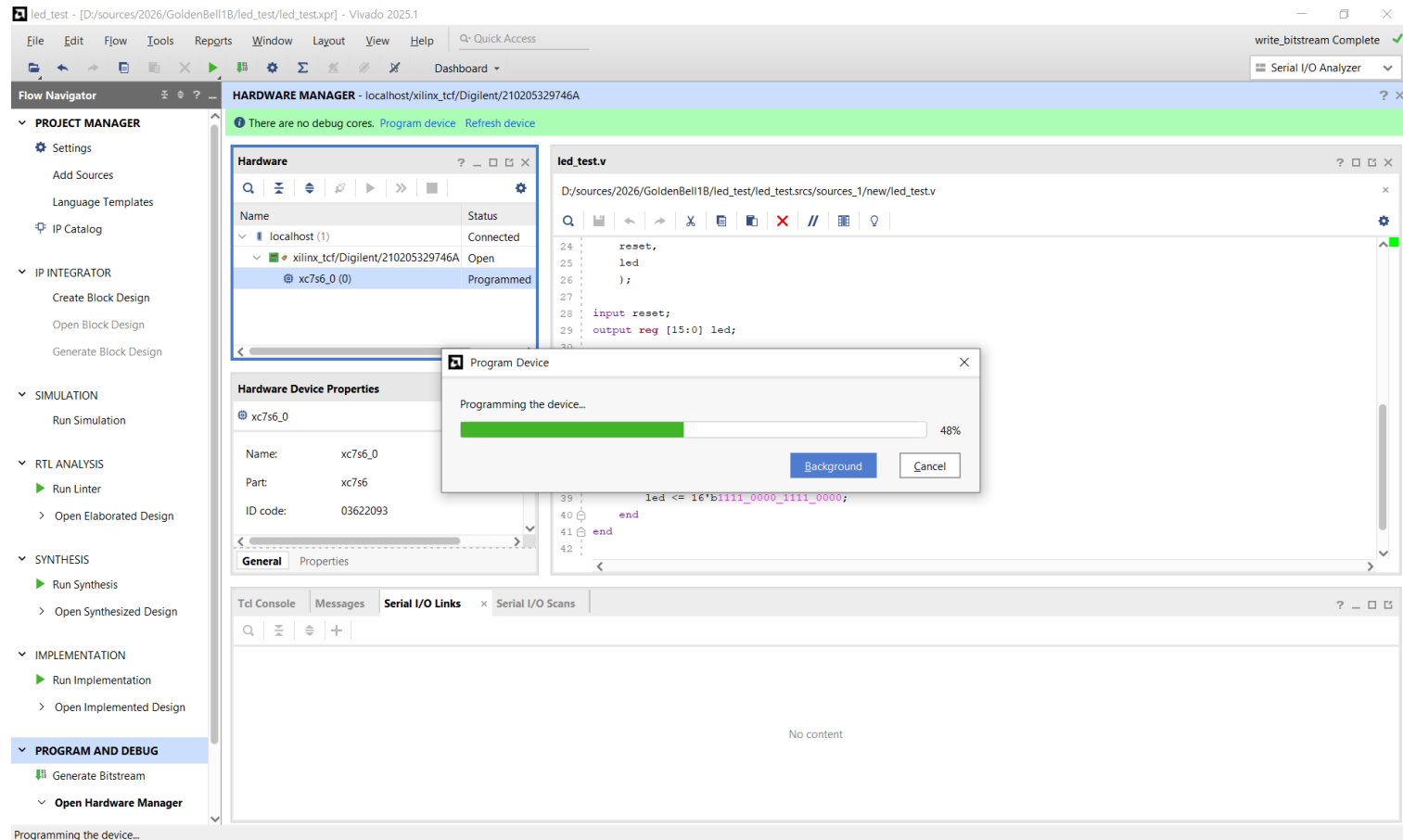
```

③ Jtag 연결이 정상적이면 device가 창에 뜸

Tcl Console Messages Serial I/O Links Serial I/O Scans

No content



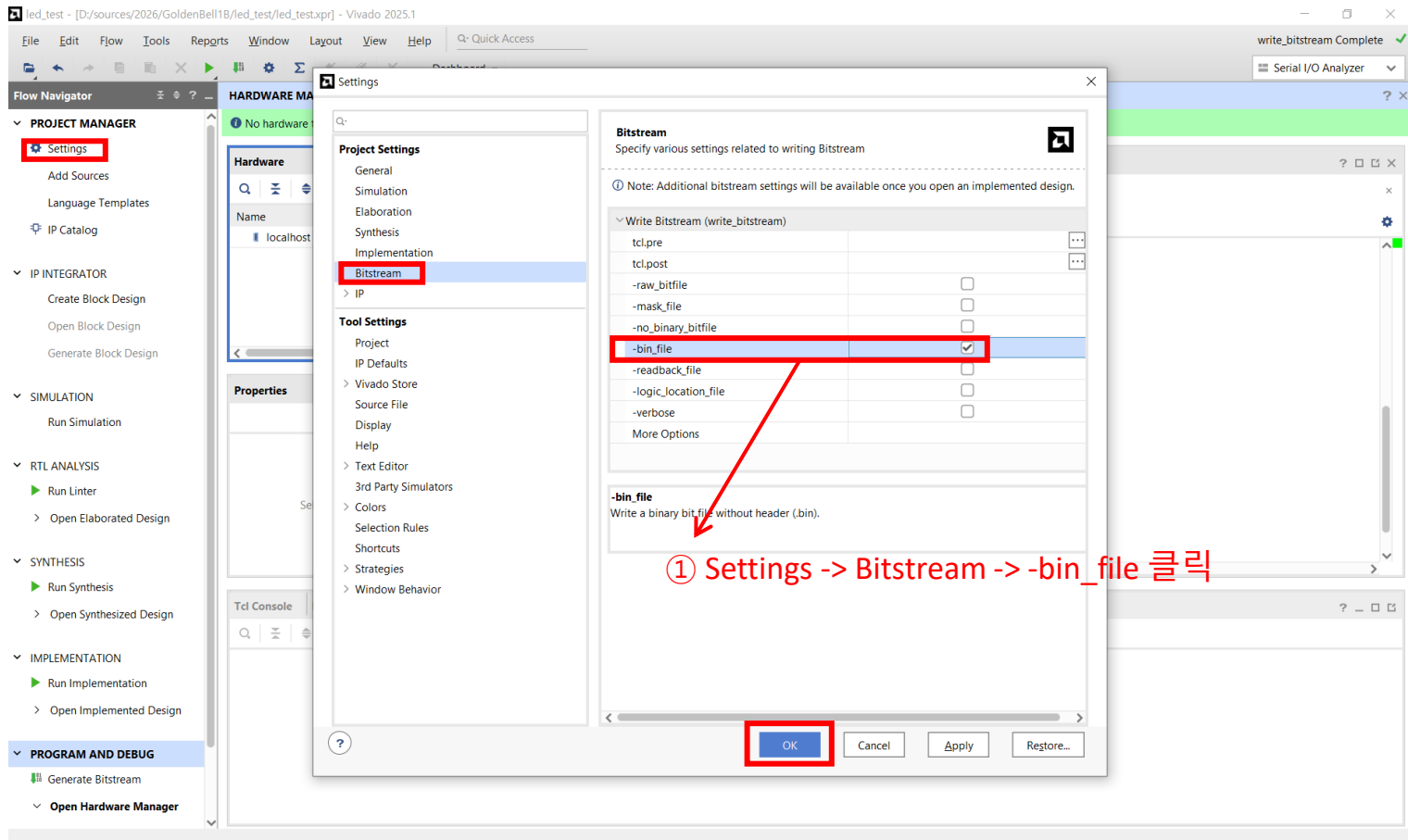


⑤ Bit 파일을 프로그램함

SPI Flash에 FPGA Bit File 프로그램

SPI Flash

- 최초 전원 인가 시 바로 SPI flash에 저장된 bit file 실행
- SPI flash spec
 - Macronix 16Mb NOR Flash memory (mx25v1635f)
 - Datasheet는 <GoldenBell1B_CPU_test_platform_manual/datasheet> 폴더에 존재
- SPI flash에 program하는 방법
 - BIN file 생성
 - Add Configuration Memory Device에 SPI flash를 추가
 - BIN file programming



led_test - [D:/sources/2026/GoldenBell1B/led_test/led_test.xpr] - Vivado 2025.1

File Edit Flow Tools Reports Window Layout View Help Q: Quick Access

Synthesis and Implementation Out-of-date details

Default Layout

Flow Navigator PROJECT MANAGER - led_test

PROJECT MANAGER

- Settings
- Add Sources
- Language Templates
- IP Catalog

IP INTEGRATOR

- Create Block Design
- Open Block Design
- Generate Block Design

SIMULATION

- Run Simulation

RTL ANALYSIS

- Run Linter
- Open Elaborated Design

SYNTHESIS

- Run Synthesis
- Open Synthesized Design

IMPLEMENTATION

- Run Implementation
- Open Implemented Design

PROGRAM AND DEBUG

- Generate Bitstream**
- Open Hardware Manager

PROJECT MANAGER - led_test

Sources

- Design Sources (1)
 - led_test (led_test.v)
- Constraints (1)
 - constrs_1 (1)
 - example.xdc
- Simulation Sources (1)
- Utility Sources (1)

Hierarchy Libraries Compile Order

Source File Properties

example.xdc

Enabled

Location: D:/sources/2026/GoldenBell1B/led_test

Type: XDC

Size: 75 KB

General Properties

Project Summary led_test.v

D:/sources/2026/GoldenBell1B/led_test/srcs/sources_1/new/led_test.v

```

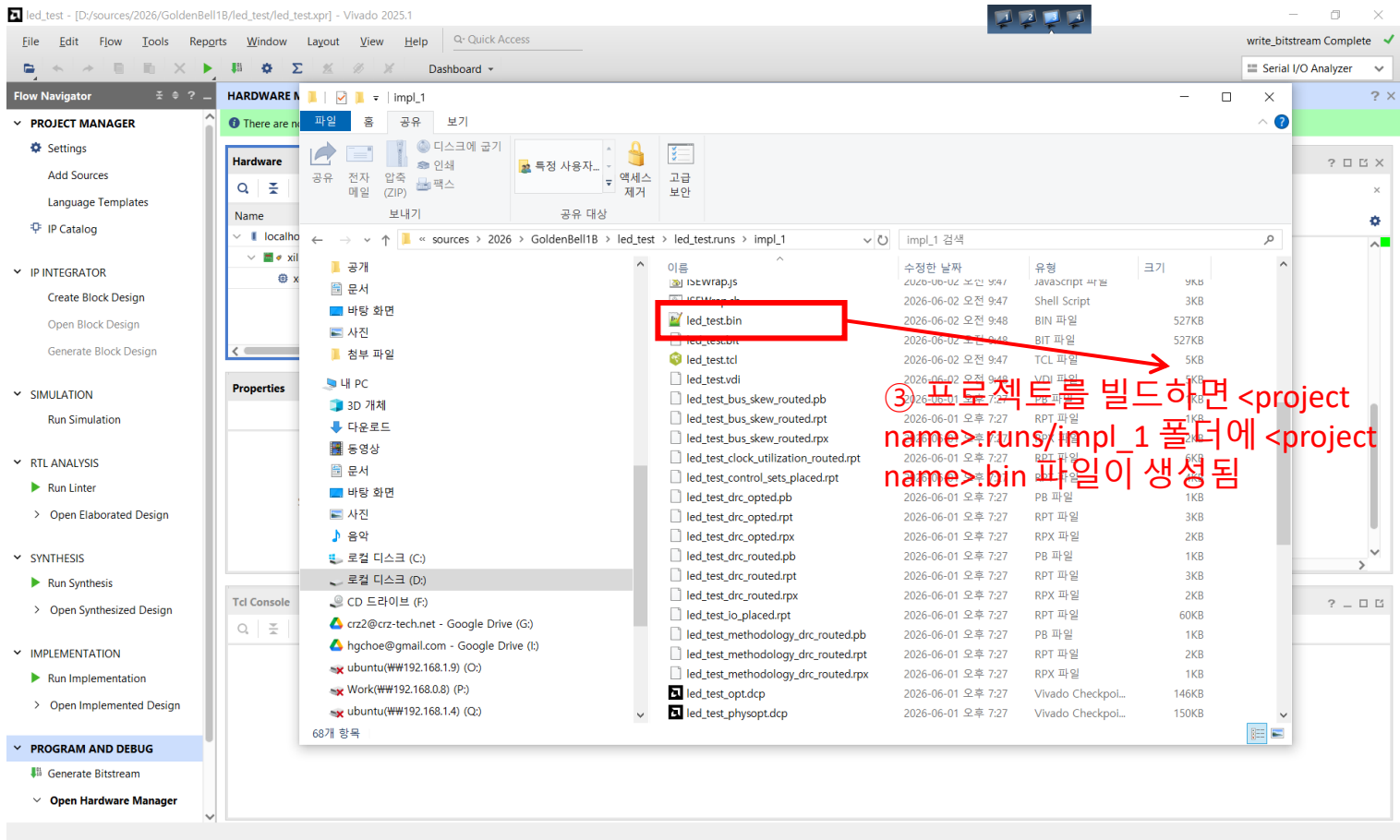
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module led_test(
24     reset,
25     led
26 );
27
28 input reset;
29 output reg [15:0] led;
30
31 always @(posedge reset)
32     led <- 0;
33 if (!reset)
  
```

Tcl Console Messages Log Reports Design Runs

Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodology	RQA Score	QoR Suggestions	LUT	FF	BRAM	URAM	DSP
synth_1	constrs_1	Synthesis Out-of-date												0	0	0	0	0
impl_1	constrs_1	Implementation Out-of-date	NA	NA	NA	NA		NA	5.880	0				0	0	0	0	0

Generate a programming file after implementation

② Generate Bitstream을 클릭하면 BIN파일이 생성됨



led_test - [D:/sources/2026/GoldenBell1B/led_test/led_test.xpr] - Vivado 2025.1

File Edit Flow Tools Repgrts Window Layout View Help Q: Quick Access write_bitstream Complete ✓ Default Layout

Flow Navigator PROJECT MANAGER - led_test

PROJECT MANAGER

- Settings
- Add Sources
- Language Templates
- IP Catalog

IP INTEGRATOR

- Create Block Design
- Open Block Design
- Generate Block Design

SIMULATION

- Run Simulation

RTL ANALYSIS

- Run Linter
- Open Elaborated Design

SYNTHESIS

- Run Synthesis
- Open Synthesized Design

IMPLEMENTATION

- Run Implementation
- Open Implemented Design

PROGRAM AND DEBUG

- Generate Bitstream
- Open Hardware Manager**

PROJECT MANAGER - led_test

Sources

- Design Sources (1)
 - led_test (led_test.v)
- Constraints (1)
 - constrs_1 (1)
 - example.xdc
- Simulation Sources (1)
- Utility Sources (1)

Source File Properties

example.xdc

☒ Enabled

Location: D:/sources/2026/GoldenBell1B/led_test

Type: XDC

Size: 1.8 KB

General Properties

Project Summary led_test.v

D:/sources/2026/GoldenBell1B/led_test/srcs/sources_1/new/led_test.v

```

24     reset,
25     led
26 );
27
28 input reset;
29 output reg [15:0] led;
30
31 always @(*)
32 begin
33 if(!reset)
34 begin
35     led <= 16'b0;
36 end
37 else
38 begin
39     led <= 16'b1111_0000_1111_0000;
40 end
41 end
42
43 endmodule
44

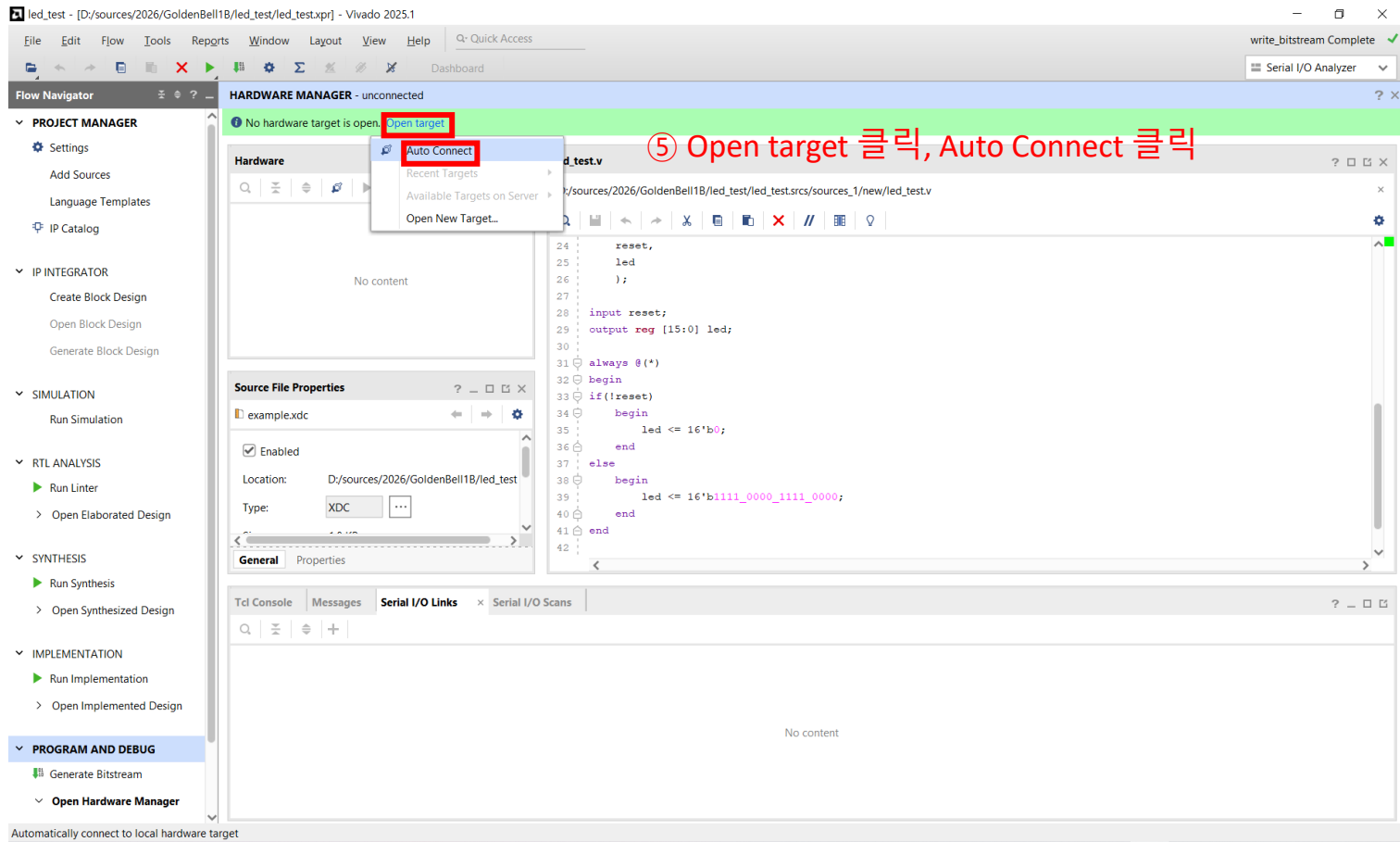
```

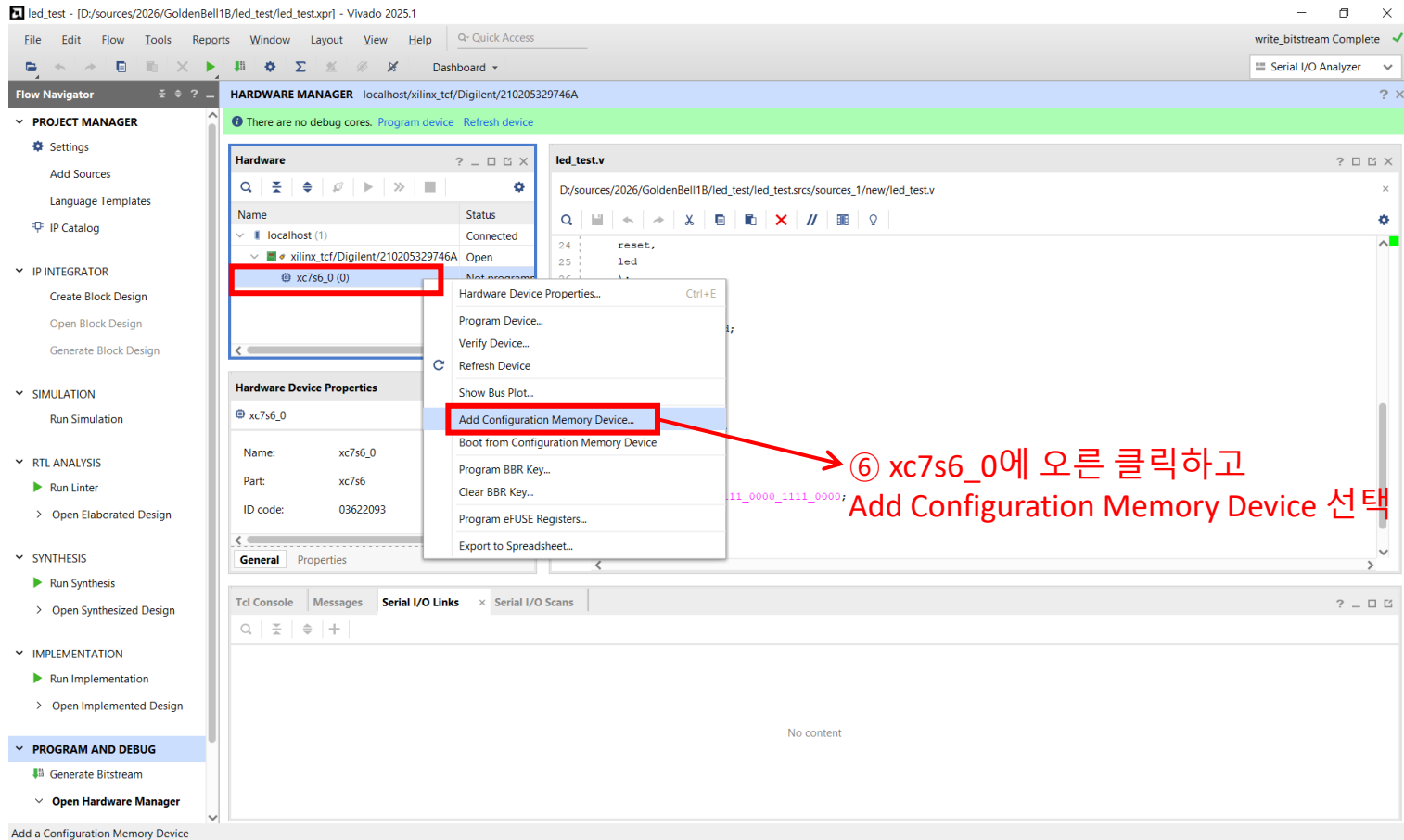
Design Runs

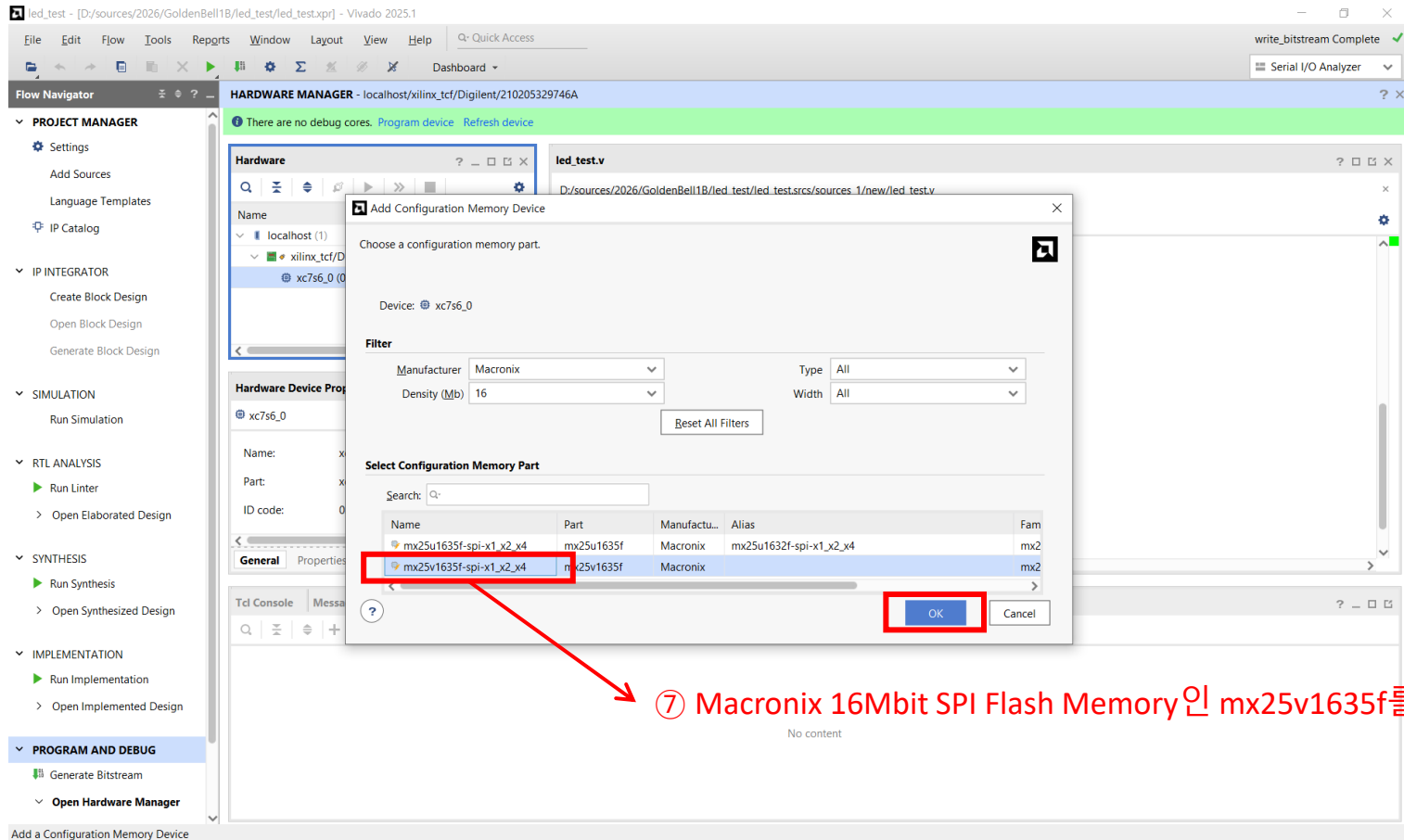
Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodology	RQA Score	QoR Suggestions	LUT	FF	BRAM	URAM	DSP
✓ synth_1	constrs_1	synth_design Complete!												0	0	0	0	0
✓ impl_1	constrs_1	write_bitstream Complete!	NA	NA	NA	NA		NA	5.880	0				0	0	0	0	0

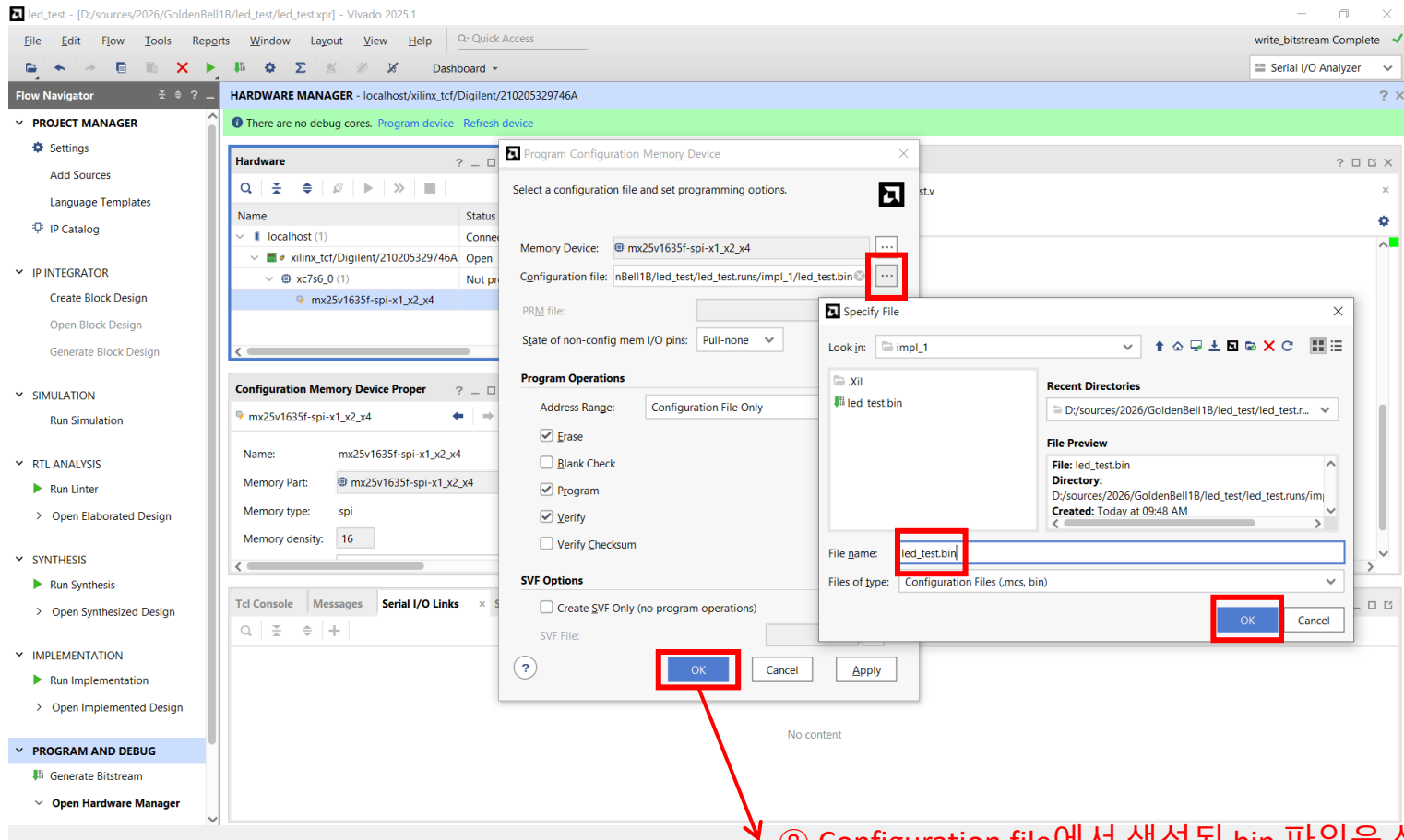
④ Open Hardware Manager를 클릭

Open the hardware program and debug manager

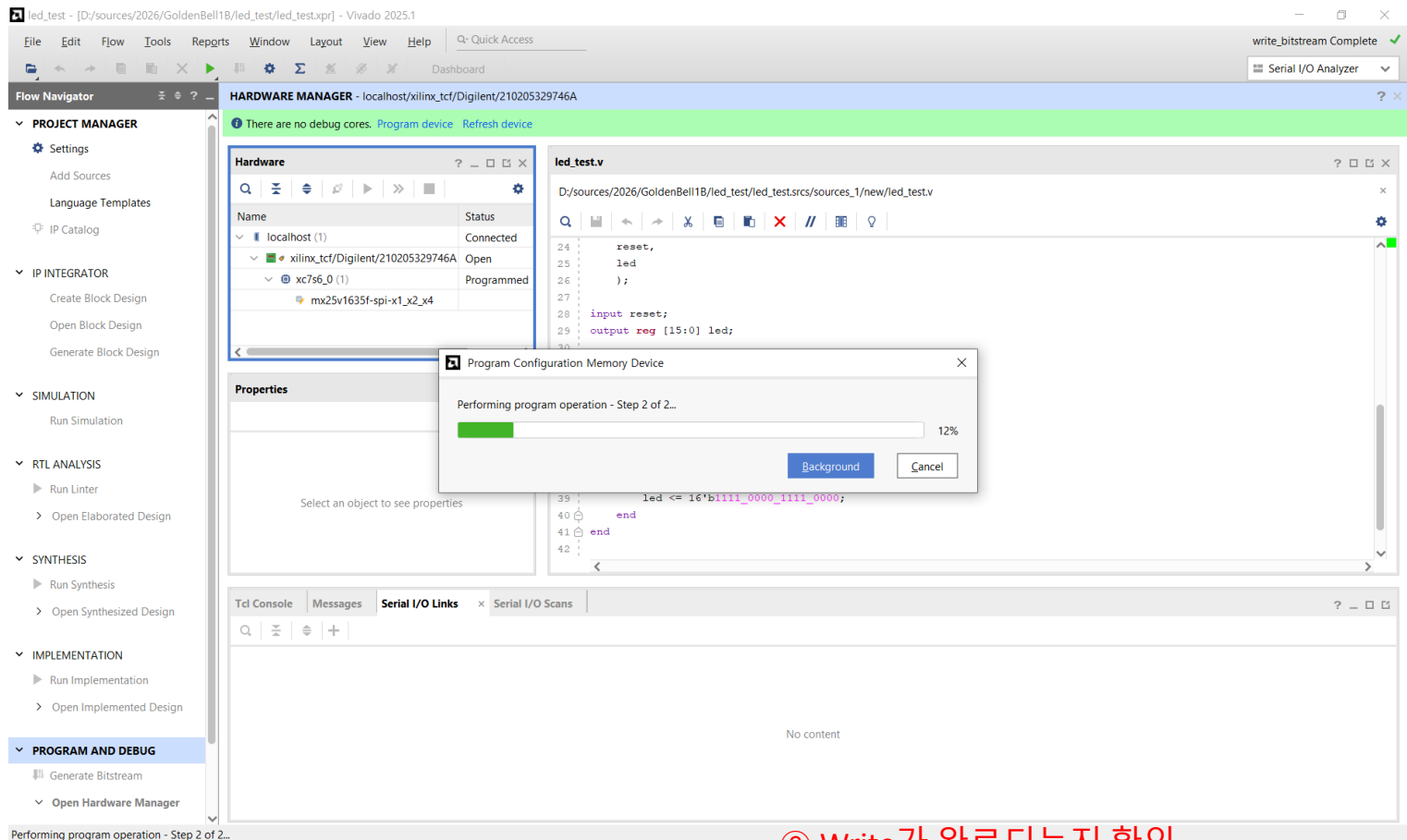








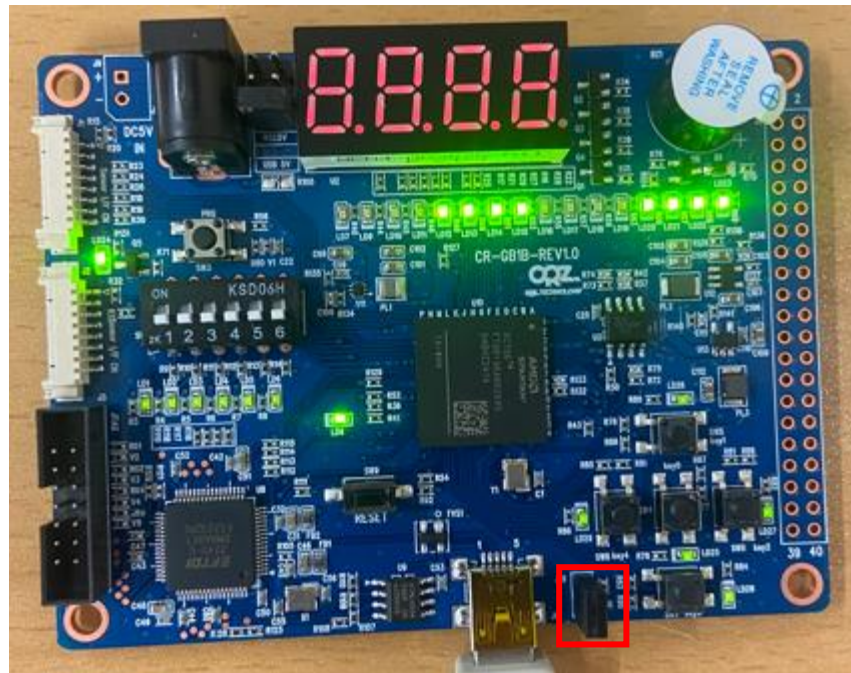
⑧ Configuration file에서 생성된 bin 파일을 선택



⑨ Write가 완료되는지 확인

SPI Flash 동작 확인

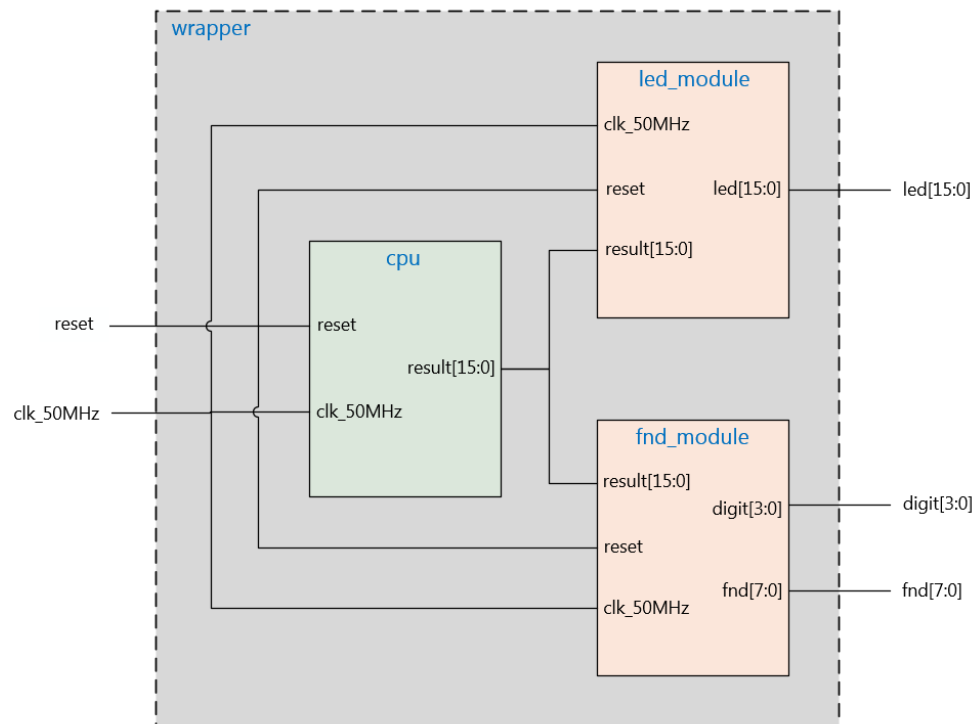
- SPI Flash로 Configuration하기 위해서는 J10을 jumper 로 연결
- USB cable을 탈착하여 보드에 다시 전원을 인가



⑩ J10을 점퍼로 연결

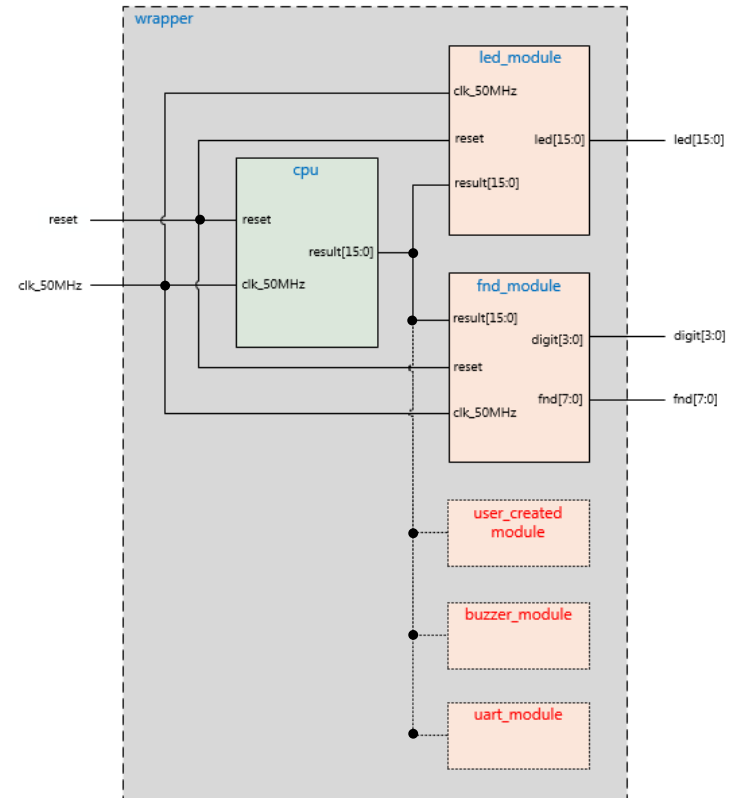
CPU Design Example

Basic Block Design



Purpose

- Verilog 언어를 사용하여 CPU를 구현하고, 이를 시뮬레이션으로 검증한 다음, 실제 FPGA 보드상에서 동작 시켜, 그 결과를 가시적으로 확인
- 실제로 확인하고 싶은 값을 CPU의 output으로 설정, 이를 FND와 LED로 확인해 볼 수 있음
- I2C / UART / Buzzer 등 추가 모듈을 통해 다양한 실험 가능



Test Flow

- CPU 모듈의 output port인 result가 led_module & fnd_module의 input result가 wire로 연결 되어있음
- CPU 모듈을 각자 구현한 CPU 모듈로 바꾼 뒤 확인하고 싶은 결과를 result에 연결
- Bit-stream 생성 후 FPGA에 program 후 실행

첨부자료

- Example_test_project
 - CPU project 결과 검증: test_platform
 - LED 예제: led_test
 - FND 예제: fnd_test
 - KEY 예제: key_test
 - Dip switch 예제: dipSwitch_test
 - Buzzer 예제: buzzer_test
 - Advanced Buzzer 예제: buzzer_ad_test
 - UART 예제: uart_test
 - Pin-map: example.xdc
- Vivado 실습 소스 위치
 - Google Drive : [vivado source#1](#)
 - Github : [vivado source#2](#)

Thank You